Composites

MO-201 Materials Science (Odd Sem 2026-27)

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Reference Textbook: Fundamentals of Materials Science by W.D. Callister

What are Composites?

- *A composite is composed of two (or more) individual materials that come from the well-known categories of materials — metals, ceramics, and polymers*

- The objective is to
  - Achieve a combination of properties that is not displayed by any single material
  - Incorporate the best characteristics of each of the component materials

- Naturally occurring composites
  - Wood consists of cellulose fibers in lignin
  - Bone consists of a hard but brittle material called hydroxyapatite (predominantly composed of calcium phosphate) and a soft and flexible material called collagen (a protein type)

- Fiberglass
  - Small glass fibers embedded within a polymeric material (epoxy/polyester)
  - Glass fibers are relatively strong and stiff (but also brittle), whereas the polymer is more flexible
  - Thus, fiberglass is relatively stiff, strong and flexible; also, it has a low density

A few more examples

- Carbon fiber–reinforced polymer (CFRP)
  - Stiffer and stronger than glass fiber–reinforced materials but more expensive
  - Aircraft and aerospace applications, high-tech sporting equipment (e.g., bicycles, golf clubs, tennis rackets, skis/snowboards) and recently in automobile bumpers

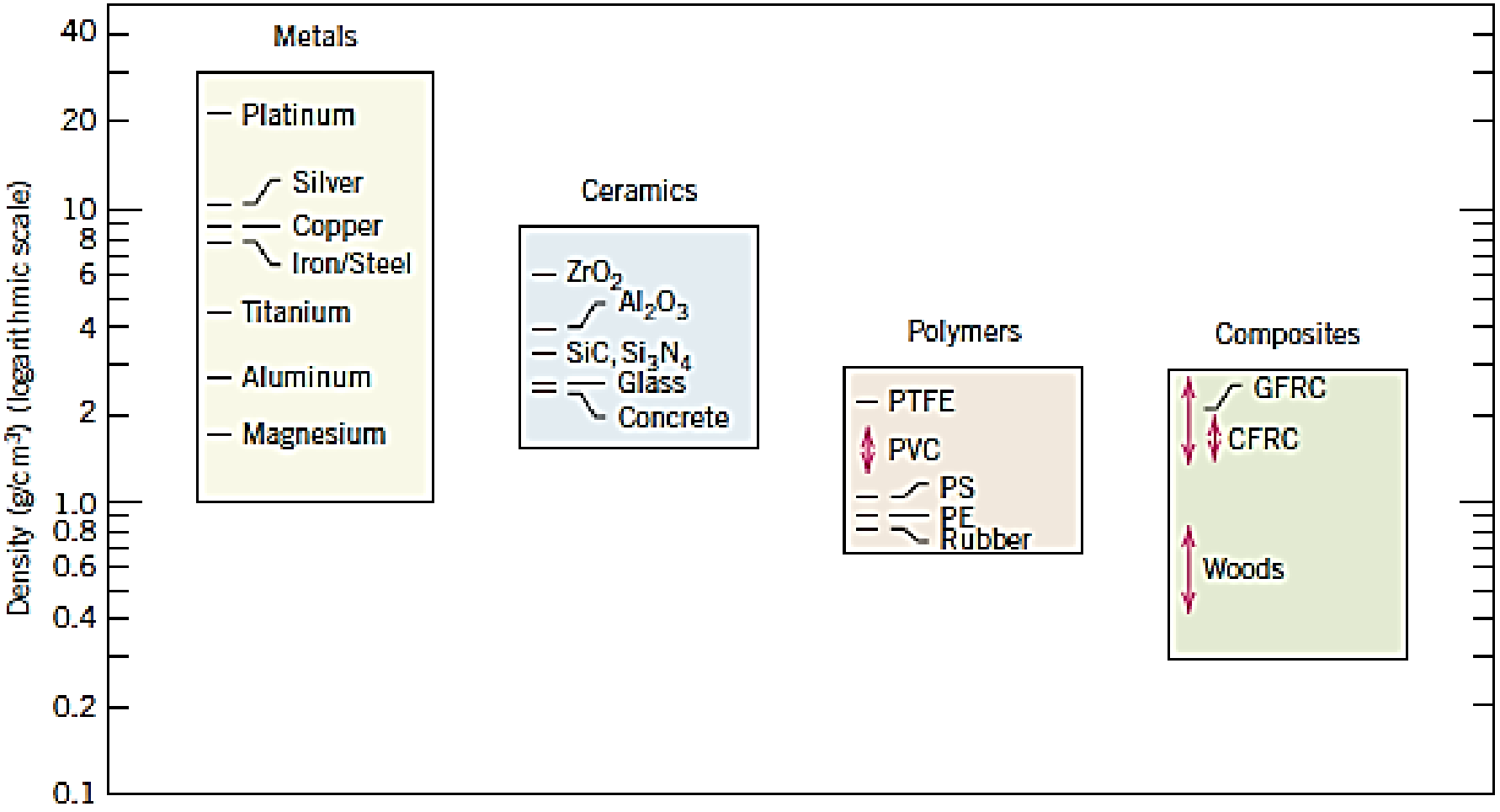
- Al-B composite
  - B → low density and high elastic modulus (E = 440 GN m^{-2}), but brittle
  - B used as a reinforcing fibre for a ductile matrix such as Al
  - Elastic modulus increased due to the presence of the B fibers
  - The ductile matrix stops a propagating crack if a fiber embedded in it breaks accidentally
  - If the entire material were to consist of B only, a propagating crack would culminate in the fracture of the entire cross-section

- Thoria dispersed Ni - the high-temperature strength of nickel alloys may be enhanced significantly by the addition of about 3 vol. % thoria (ThO_2) as finely dispersed particles

- Concrete is a common large-particle composite
  - A composite material consisting of an aggregate of sand/gravel particles that are bound together in a solid body by some type of binding medium (cement)

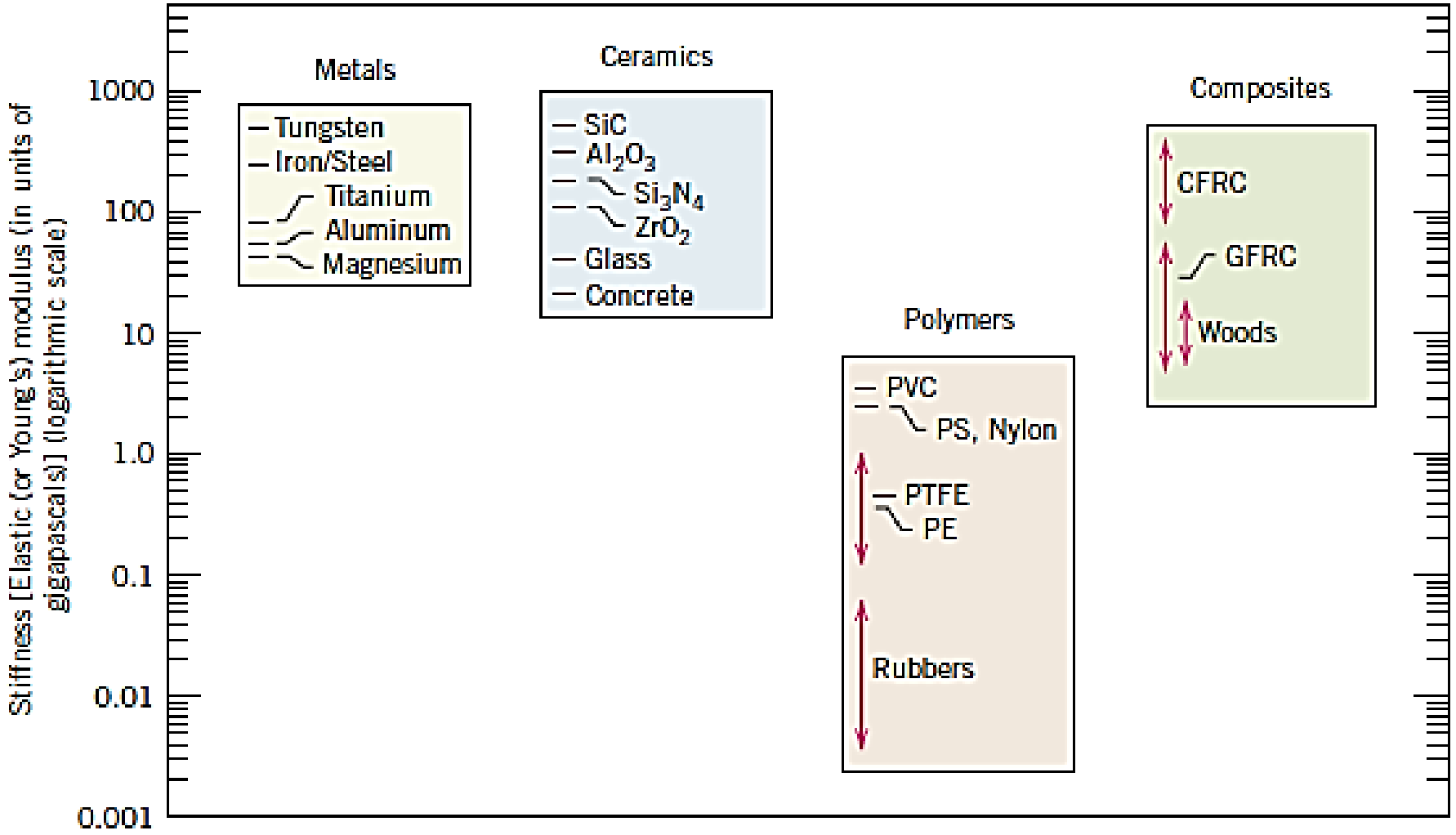
Where do Composites stand?

Room temperature density



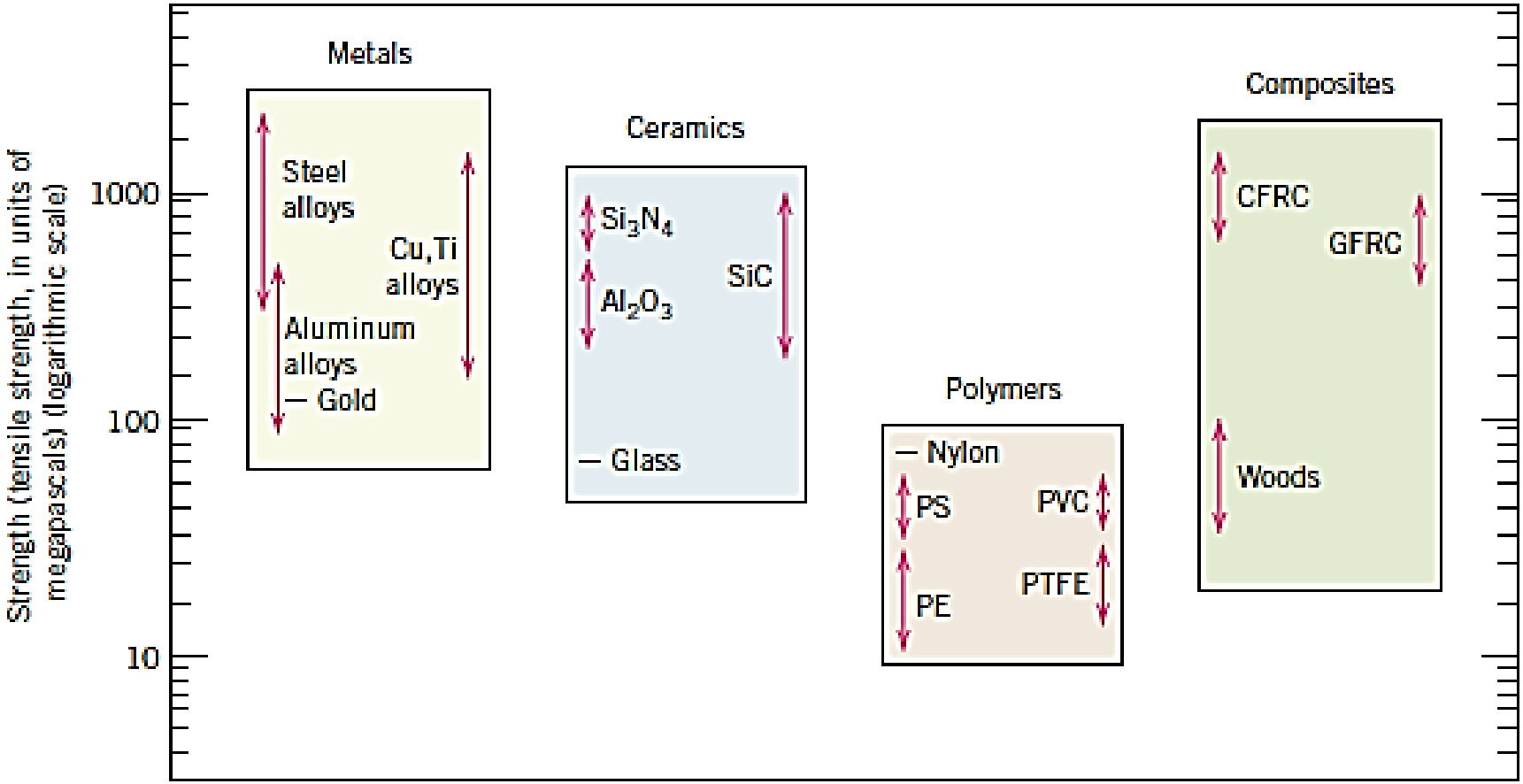
Where do Composites stand?

Room temperature stiffness



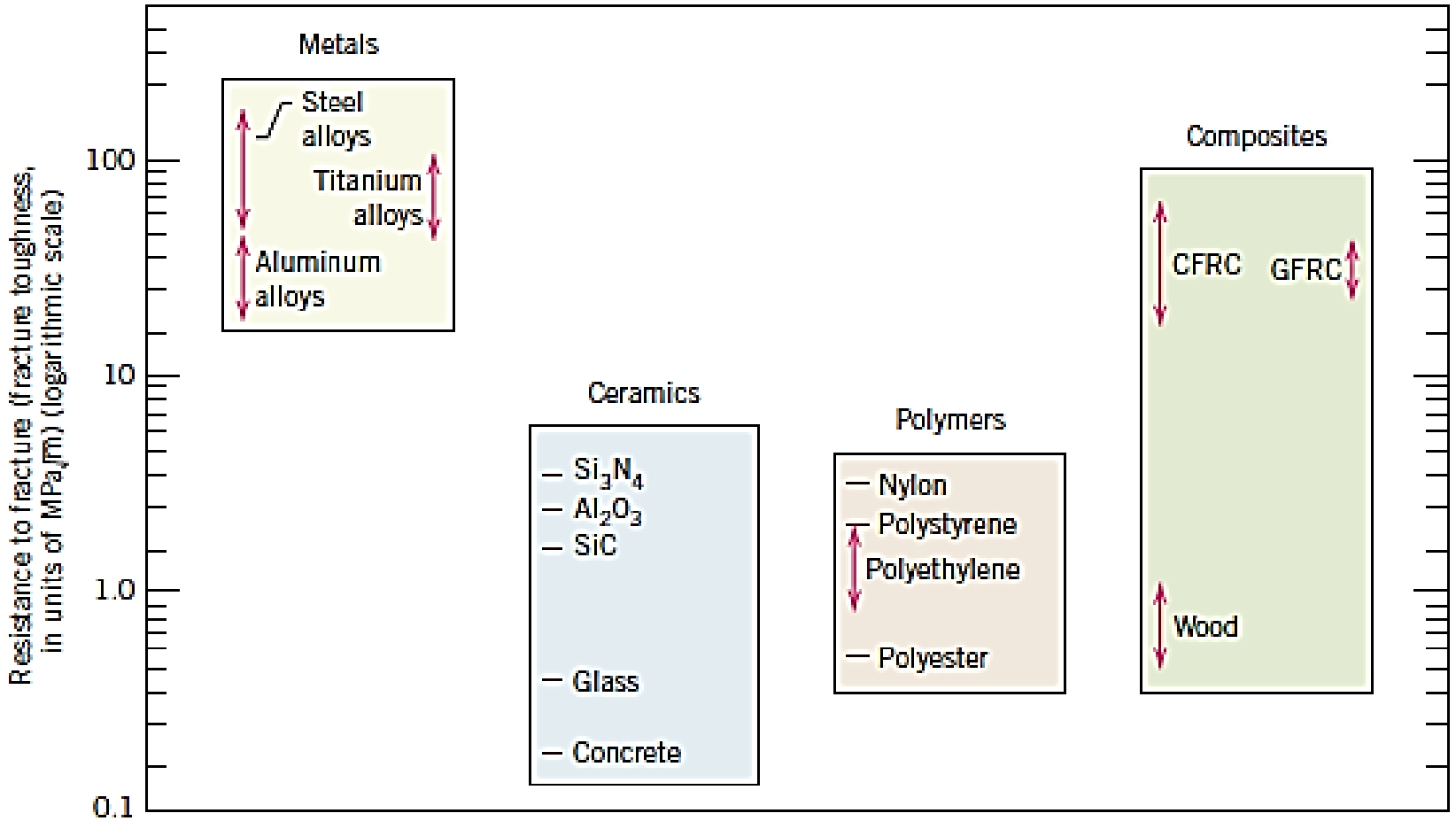
Where do Composites stand?

Room temperature tensile strength



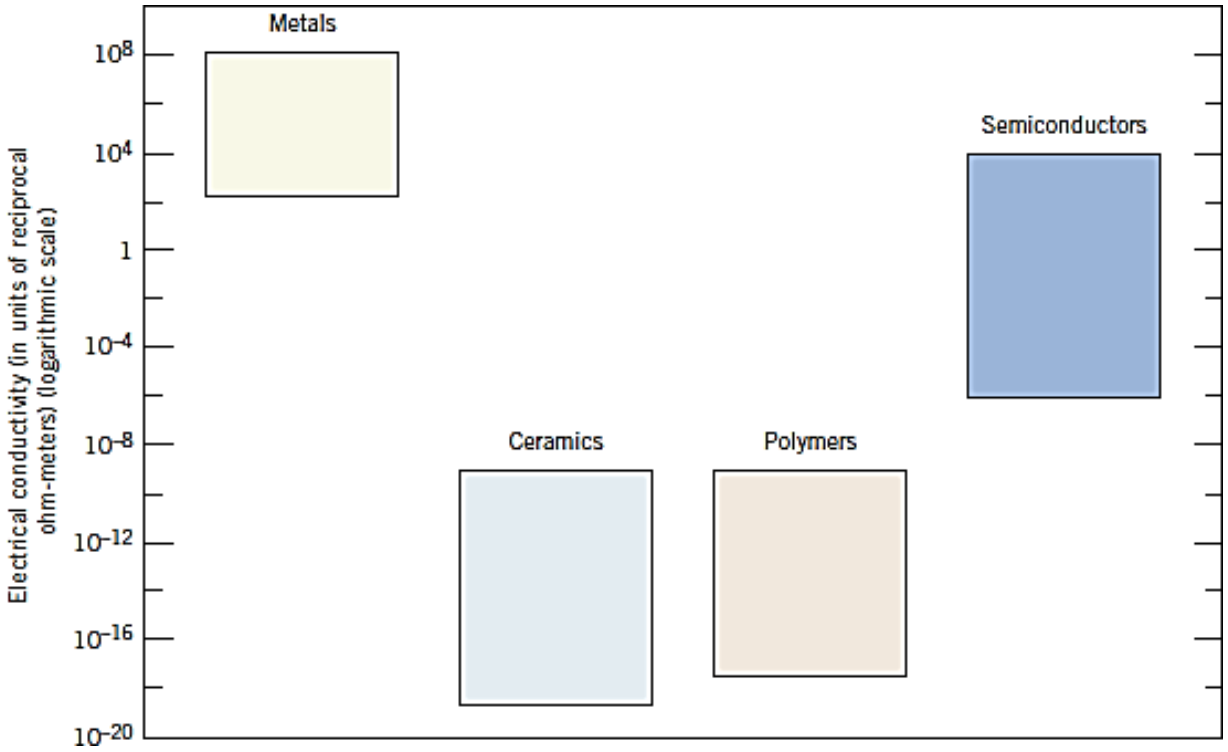
Where do Composites stand?

Room temperature fracture toughness



Where do Composites stand?

Room temperature electrical conductivity



Examples of semiconductor based composite materials

composite[a]	preparative route
Ni/TiO_2	electrodeposition
Ni/CdS	electrodeposition
Pt/TiO_2	electrodeposition
Zn/TiO_2	electrodeposition
polypyrrole/TiO_2	electrodeposition
phthalocyanine/ZnO	electrodeposition
polypyrrole/γ-Fe_2O_3	electrodeposition
phthalocyanine/TiO_2	chemical
polyaniline/γ-Fe_2O_3	chemical
Nafion/CdS	chemical
Nafion/ZnS	chemical
polypyrrole/SnO_2	chemical
Nafion/TiO_2	chemical
clay/TiO_2	chemical
clay/Fe_2O_3	chemical
activated carbon/TiO_2	chemical
alumina/TiO_2	chemical
silica/TiO_2	chemical
zeolite/TiO_2	chemical
zirconia/TiO_2	chemical
glass microbubble/TiO_2	thermolytic
fly ash/TiO_2	thermolytic
alumina/TiO_2	MOCVD

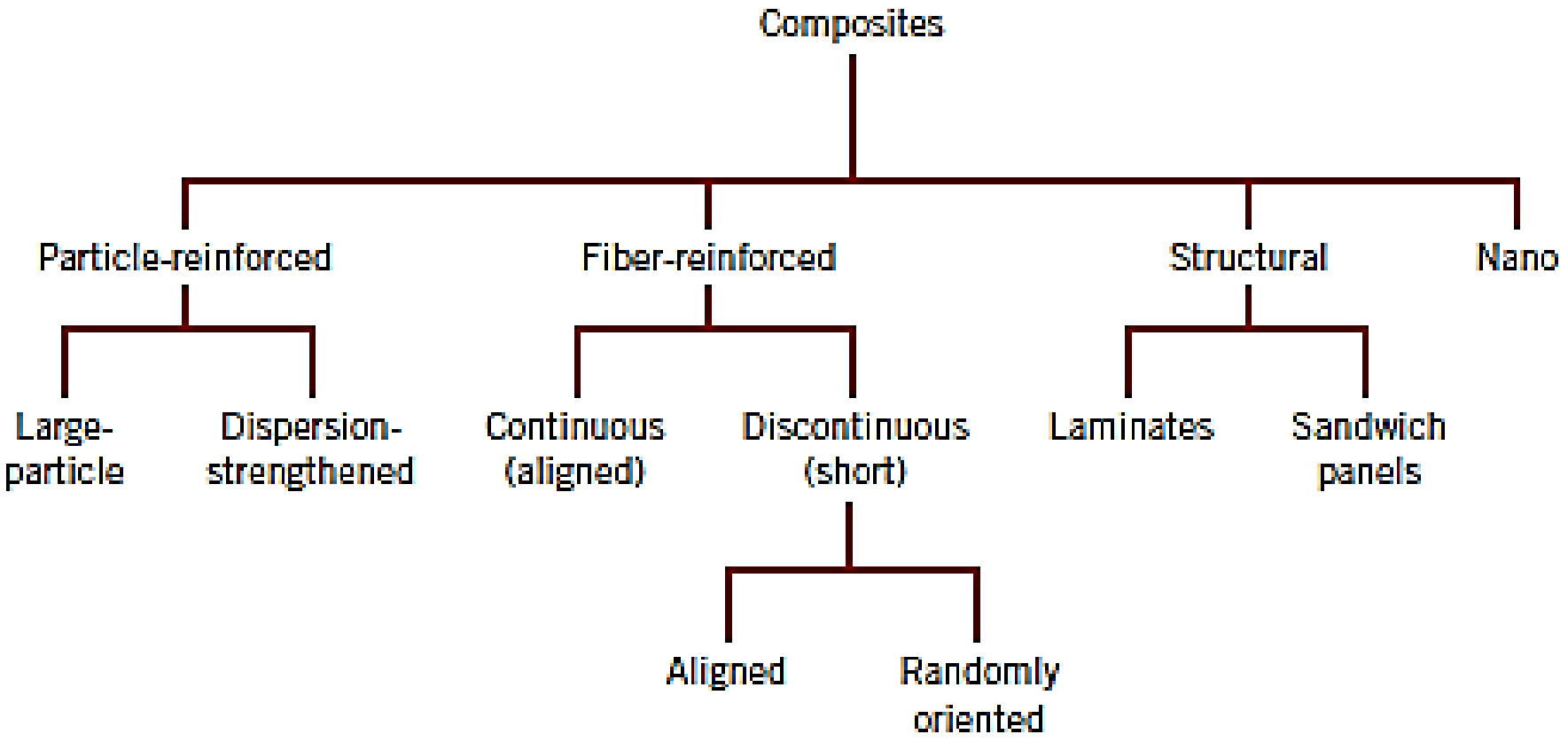
[a]The composite is designated in each case by the matrix first followed by the dispersed semiconductor phase

Other Examples of Semiconductor/ Semiconductor Composites

CdS/TiO_2
CdS/ZnO
Cd_3P_2/TiO_2
Cd_3P_2/ZnO
AgI/Ag_2S
Ag_2S/CdS
PbS/TiO_2
WO_3/WS_2
ZnS/ZnO
ZnSe/ZnO
CdS/AgI
CdS/CdSe
ZnS/CdS
TiO_2/CdSe
TiO_2/PbSe
TiO_2/Fe_2O_3
LiO/TiO_2
WO_3/TiO_2
ZnO/TiO_2

K. Rajeshwar et al., Chem. Mater. 2001, 13, 2765-2782

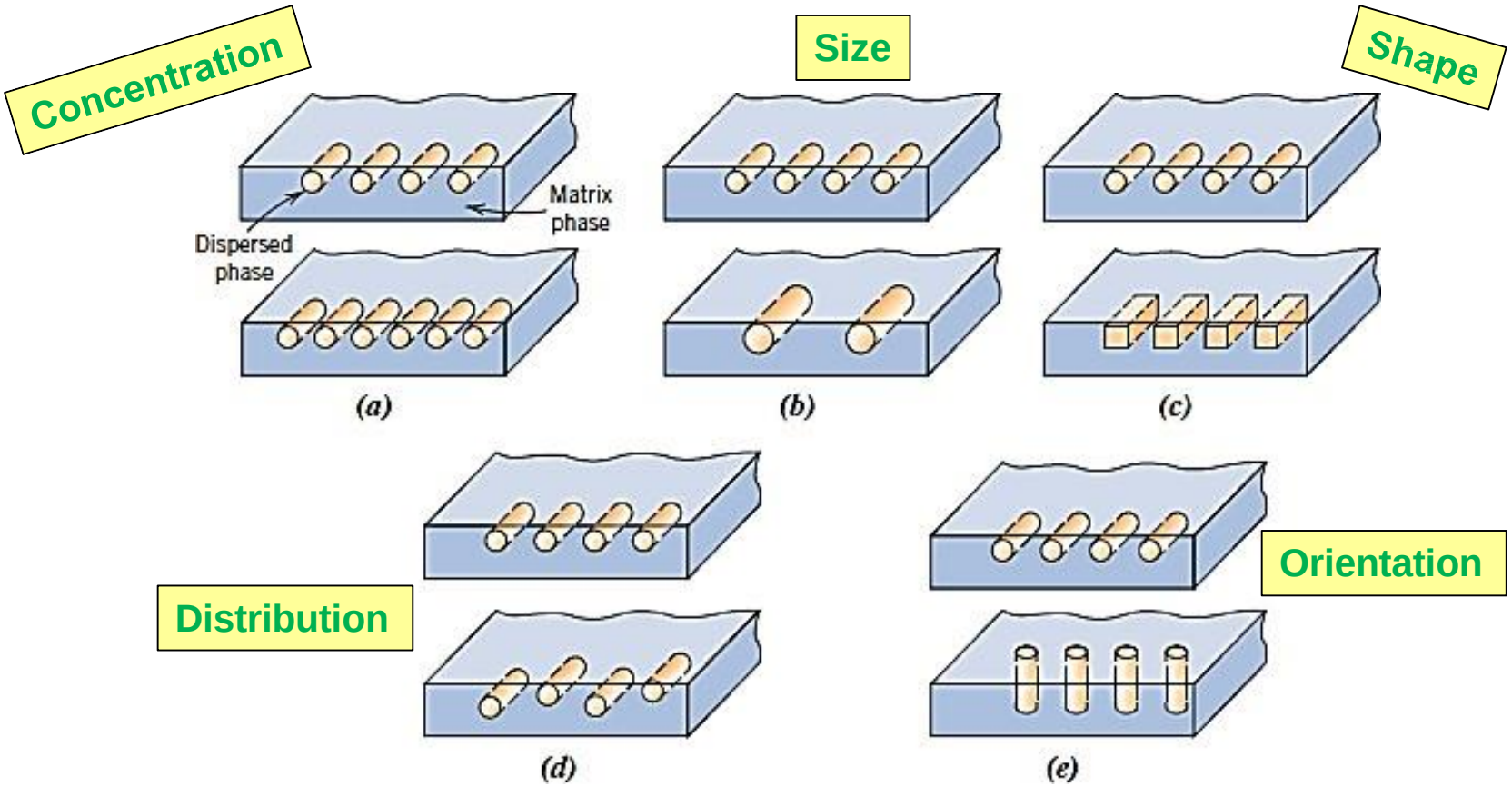
Classification of Composites



Matrix and Dispersed Phases

- Many composite materials are composed of just two phases; one is termed the ***matrix***, which is continuous and surrounds the other phase, often called the ***dispersed phase***

- The properties of composites are a function of the properties of the constituent phases, their relative amounts, and the geometry of the dispersed phase



Types of Composites (contd.)

- Some broad differences between particle-reinforced, fiber-reinforced, structural and nano composites are as follows

- The dispersed phase for particle-reinforced composites is equiaxed (i.e., particle dimensions are approximately the same in all directions)

- For fiber-reinforced composites, the dispersed phase has the geometry of a fiber (i.e., a large length-to-diameter ratio)

- Structural composites are multi-layered and designed to have low densities and high degrees of structural integrity

- For nanocomposites, dimensions of the dispersed phase particles are on the order of nanometers

Particle-reinforced Composites

- Particle-reinforced composites are further classified into ***large-particle*** and ***dispersion-strengthened***, based on reinforcement/strengthening mechanisms

- The term large is used to indicate that particle–matrix interactions cannot be treated on the atomic or molecular level; rather, continuum mechanics is used.

- For most of these composites, the particulate phase is harder and stiffer than the matrix; these reinforcing particles tend to restrain movement of the matrix phase in the vicinity of each particle

- In essence, the matrix transfers some of the applied stress to the particles, which bear a fraction of the load

- The degree of reinforcement or improvement of mechanical behavior depends on strong bonding at the matrix–particle interface

Particle-reinforced Composites

- For dispersion-strengthened composites, particles are normally much smaller, with diameters between 0.01 and 0.1 μm (10 and 100 nm)
- Particle–matrix interactions that lead to strengthening occur on the atomic or molecular level
- Whereas the matrix bears the major portion of an applied load, the small dispersed particles hinder or impede the motion of dislocations
- Thus, plastic deformation is restricted, resulting in increase in yield and tensile strengths, as well as hardness

Rule of mixtures

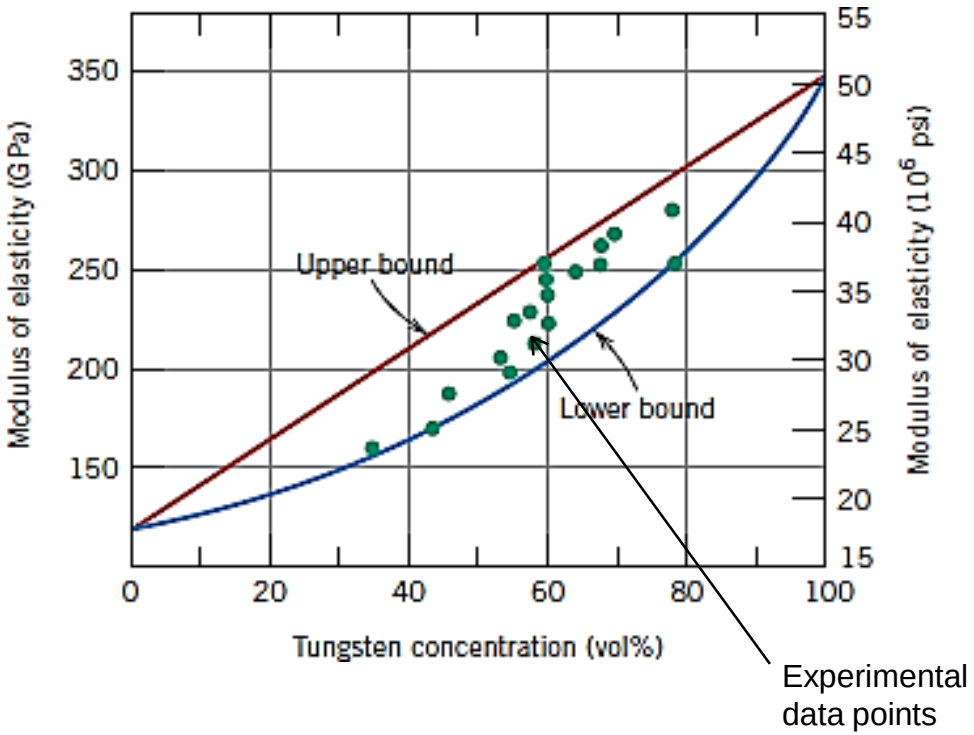
Modulus of elasticity upper bound equation for 2-phase composite

$$E_c(u) = E_m V_m + E_p V_p$$

Modulus of elasticity lower bound equation for 2-phase composite

$$E_c(l) = \frac{E_m E_p}{V_m E_p + V_p E_m}$$

Upper and lower bound for Copper-Tungsten composite where tungsten is the particle phase



E and *V* denote the elastic modulus and volume fraction, respectively, and the subscripts *c*, *m*, and *p* represent composite, matrix, and particulate phases, respectively

Cermets

- Large-particle composites are used with all three material types (metals, polymers and ceramics). The ***cermets*** are examples of ceramic–metal composites

- The most common cermet is cemented carbide, which is composed of extremely hard particles of a refractory carbide ceramic such as tungsten carbide (WC) or titanium carbide (TiC) embedded in a matrix of a metal such as cobalt or nickel

- Used extensively as cutting tools for hardened steels

- Toughness is enhanced by their inclusion in the ductile metal matrix, which isolates the carbide particles from one another and prevents particle-to-particle crack propagation

- Both matrix and particulate phases are quite refractory to the high temperatures generated by the cutting action on materials that are extremely hard

- No single material could possibly provide the combination of properties possessed by a cermet; relatively large volume fractions of the particulate phase may be used, often exceeding 90 vol. %

Carbon Black – Reinforced Rubber

- Both elastomers and plastics are frequently reinforced with various particulate materials. Use of many modern rubbers would be severely restricted without reinforcing particulate materials such as carbon black

- Carbon black consists of very small and essentially spherical particles of carbon, produced by the combustion of natural gas or oil in an atmosphere that has only a limited air supply

- When added to vulcanized rubber, this extremely inexpensive material enhances tensile strength, toughness, and tear and abrasion resistance - automobile tires contain on the order of 15 to 30 vol. % carbon black

- For the carbon black to provide significant reinforcement, the particle size must be extremely small, with diameters between 20 and 50 nm; also, the particles must be evenly distributed throughout the rubber and must form a strong adhesive bond with the rubber matrix

- Particle reinforcement using other materials (e.g., silica) is much less effective because this special interaction between the rubber molecules and particle surfaces does not exist

Concrete

- Concrete is a common large-particle composite in which both matrix and dispersed phases are ceramic materials *(Recall the distinction between concrete and cement discussed earlier)*

- The two most familiar concretes are those made with Portland and asphaltic cements, in which the aggregate is gravel and sand

- Asphaltic concrete is widely used primarily as a paving material, whereas Portland cement concrete is employed extensively as a structural building material

- Portland cement concrete is a major construction material, primarily because it can be poured in place and hardens at room temperature and even when submerged in water

Concrete

- However, Portland cement also has some limitations as a structural material

  - Like most ceramics, Portland cement concrete is relatively weak and extremely brittle; its tensile strength is approximately 1/15 to 1/10 its compressive strength

  - Also, large concrete structures can experience considerable thermal expansion and contraction with temperature fluctuations

  - In addition, water penetrates into external pores, which can cause severe cracking in cold weather as a consequence of freeze–thaw cycles

Portland Cement Concrete

- The ingredients for Portland cement concrete are Portland cement, a fine aggregate (sand), a coarse aggregate (gravel), and water
- The aggregate particles act as a filler material to reduce the overall cost of the concrete product because they are cheap, whereas cement is relatively expensive
- To achieve the optimum strength and workability of a concrete mixture, the ingredients must be added in the correct proportions
- Dense packing of the aggregate and good interfacial contact are achieved by having particles of two different sizes – fine sand particles should fill the void spaces between the gravel particles
- Typically, these aggregates constitute between 60% and 80% of the total volume

Portland Cement Concrete

- The amount of cement–water paste should be sufficient to coat all the sand and gravel particles; otherwise the cementitious bond will be incomplete

- Furthermore, all of the constituents should be thoroughly mixed - complete bonding between cement and the aggregate particles is contingent on the addition of the correct quantity of water; too little water leads to incomplete bonding, and too much results in excessive porosity; in either case, the final strength is less than the optimum

- Size distribution of the aggregates influences the amount of cement–water paste required

- Also, the surfaces should be clean and free from clay and silt, which prevent the formation of a sound bond at the particle surface

Reinforced Concrete

- The strength of Portland cement concrete may be increased by additional reinforcement - this is usually accomplished by means of steel rods, wires, bars (rebar), or mesh, which are embedded into the fresh and uncured concrete

- Steel serves as a suitable reinforcement material because
  - Coefficient of thermal expansion is nearly the same as that of concrete
  - In addition, steel is not rapidly corroded in the cement environment, and
  - A relatively strong adhesive bond is formed between it and the cured concrete

- Portland cement concrete may also be reinforced by mixing fibers of a high modulus material such as glass, steel, nylon, or polyethylene into the fresh concrete

- Care must be exercised in using this type of reinforcement because some fiber materials experience rapid deterioration when exposed to the cement environment

Prestressed Concrete

- Another reinforcement technique for strengthening concrete involves the introduction of residual compressive stresses into the structural member; the resulting material is called ***prestressed concrete***

- In one such prestressing technique, high-strength steel wires are positioned inside the empty molds and stretched with a high tensile force, which is maintained constant

- After the concrete has been placed and allowed to harden, the tension is released

- As the wires contract, they put the structure in a state of compression because the stress is transmitted to the concrete via the concrete–wire bond that is formed

- Another technique, in which stresses are applied after the concrete hardens, is appropriately called ***posttensioning***

Dispersion-Strengthened Composites

- Uniform dispersion of several volume % of fine particles of a very hard and inert material - may be metallic or non-metallic; oxide materials are often used

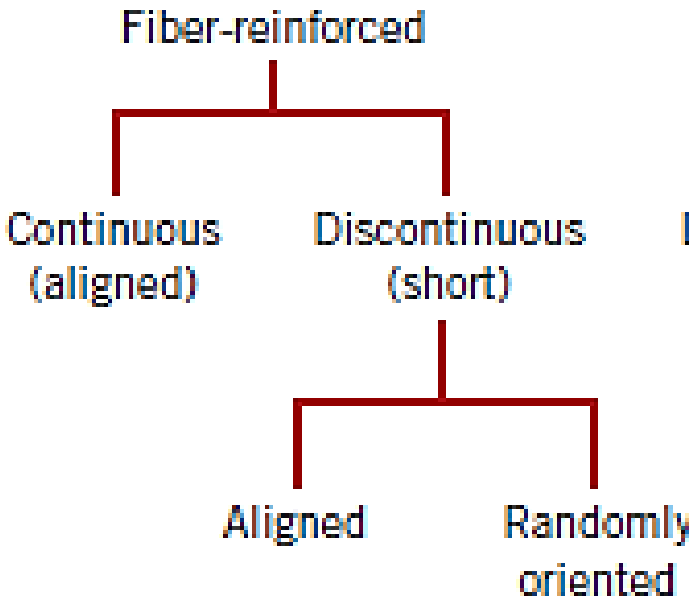
- The strengthening mechanism involves interactions between the particles and dislocations within the matrix, as with precipitation hardening

- However, the strengthening is retained at elevated temperatures and for extended time periods because the dispersed particles are chosen to be unreactive with the matrix phase; for precipitation-hardened alloys, the increase in strength may disappear upon heat treatment due to precipitate growth/dissolution

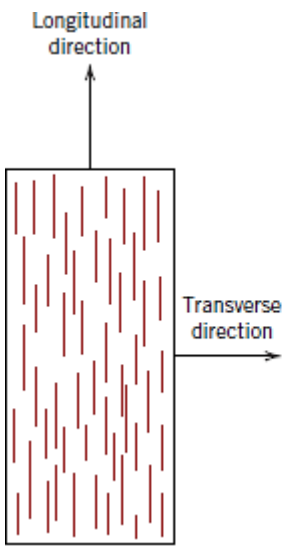
- Examples include thoria-dispersed (or TD) nickel and aluminum–aluminum oxide system (thin and adherent alumina coating on the surface of extremely small (0.1 to 0.2 μm thick) flakes of aluminum, which are dispersed within an aluminum metal matrix; this material is termed sintered aluminum powder (SAP))

Fiber-Reinforced Composites

- Fiber-reinforced composites contain the dispersed phase in the form of a fiber
- Design goals include high strength and/or stiffness on a weight basis (i.e. high specific strength and specific modulus)
- Low density fiber and matrix materials used to achieve very high specific strength and specific modulus
- Further classification of fiber-reinforced composites is based on fiber length

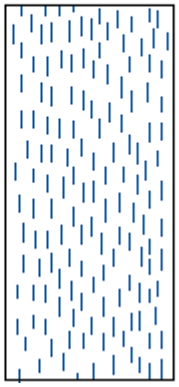


Effect of Fiber Orientation & Concentration



Continuous and aligned

Discontinuous and aligned





Discontinuous and randomly oriented

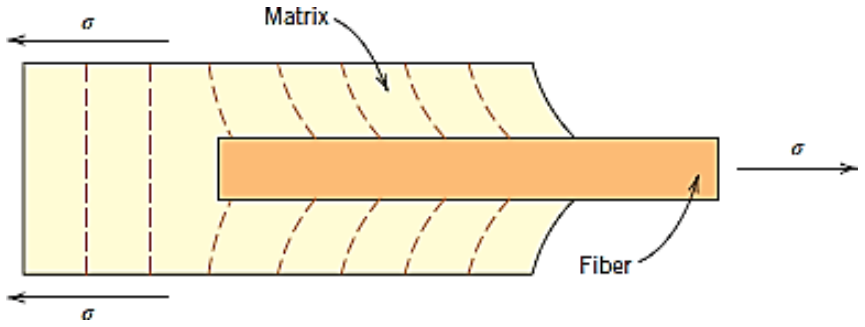
Better overall composite properties are realized when the fiber distribution is uniform

Influence of Fiber Length

- The mechanical characteristics of a fiber-reinforced composite depend on
  (1) the properties of the fiber
  (2) degree of transmittance of applied load to fibers (by matrix)

- Extent of load transmittance depends on the magnitude of the interfacial bond between the fiber and matrix phases

- Matrix deformation pattern as shown schematically in the Figure on the right (there is no load transmittance from the matrix at each fiber extremity)

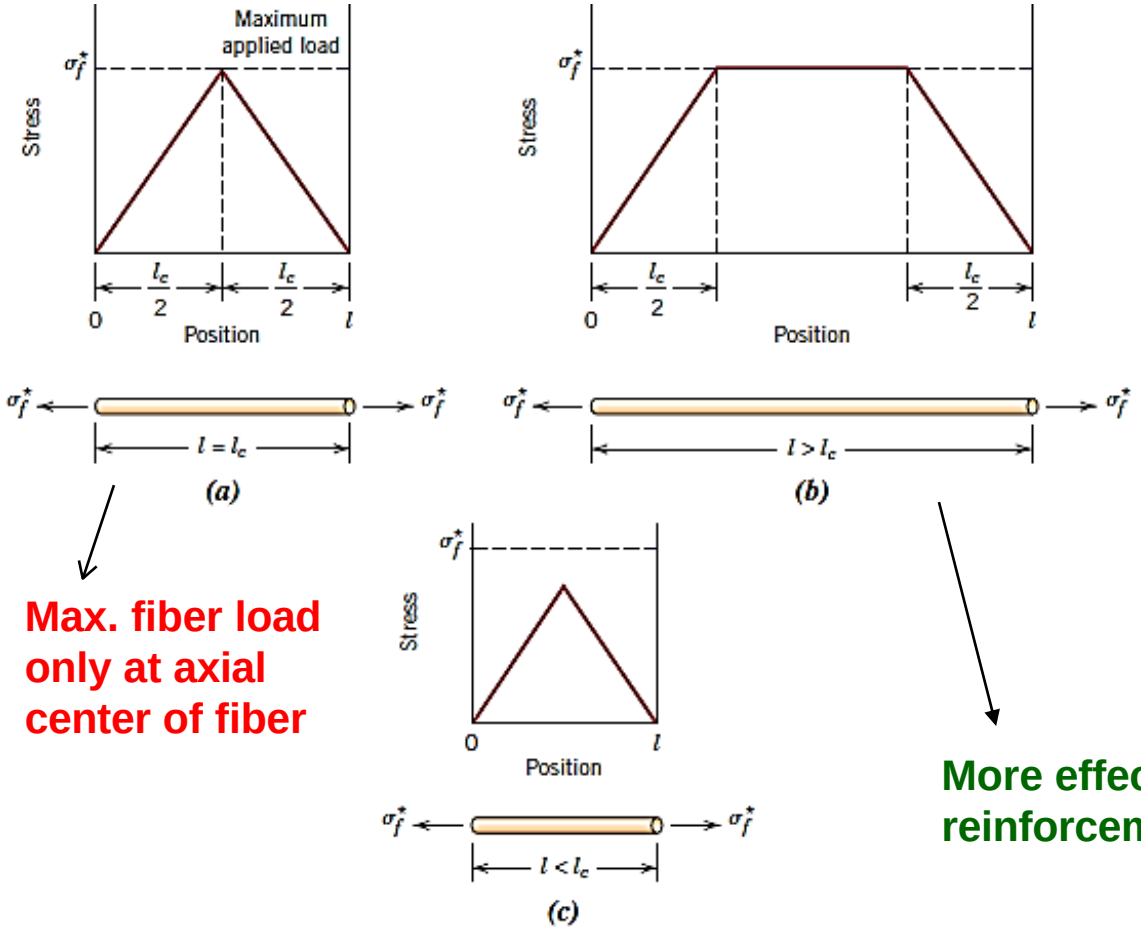


- Effective strengthening and stiffening of the composite involves a **critical fiber length**

  $l_c = \sigma_f^* d/2\tau_c$, where l_c is critical fiber length, d is fiber diameter, σ_f^* is ultimate tensile strength of the fiber, τ_c is fiber-matrix bond strength or shear yield strength of the matrix, whichever is smaller

- For a number of glass and carbon fiber–matrix combinations, this critical length is on the order of 1 mm, which ranges between 20 and 150 times the fiber diameter

Influence of Fiber Length



Stress–position profiles for a fiber–reinforced composite that is subjected to a tensile stress equal to the fiber tensile strength σ_f^* for (a) $l = l_c$ (b) $l > l_c$ (c) $l < l_c$

Max. fiber load only at axial center of fiber

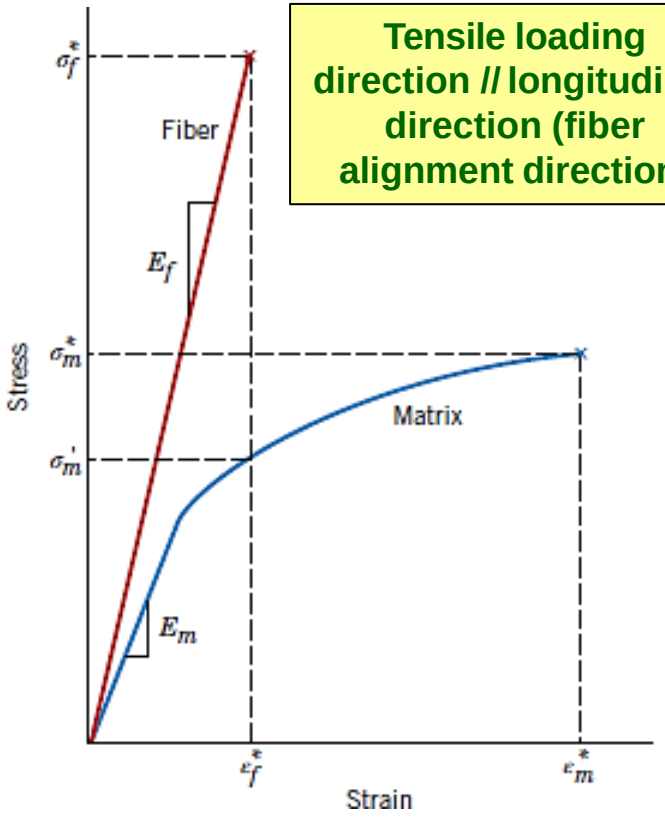
More effective fiber reinforcement

Fibers for which $l \gg l_c$ (usually $l > 15l_c$) are termed continuous; discontinuous or short fibers have lengths shorter than this.

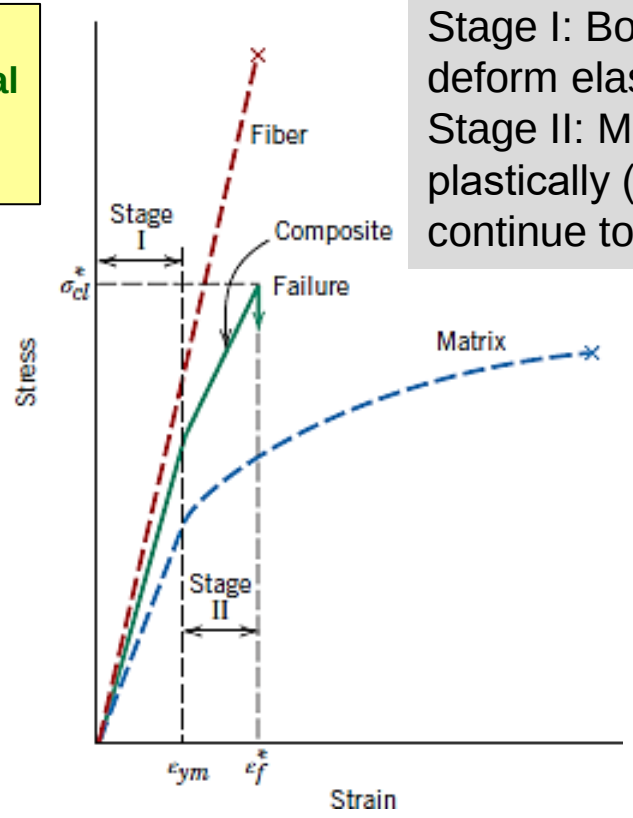
Continuous and Aligned Fiber Composites

- Mechanical responses of this type of composite depend on several factors, including the stress–strain behaviors of fiber and matrix phases, the phase volume fractions and the direction in which the stress or load is applied

- Also, the properties of a composite having its fibers aligned are highly anisotropic, that is, they depend on the direction in which they are measured



Tensile loading direction // longitudinal direction (fiber alignment direction)



Stage I: Both fibers and matrix deform elastically
Stage II: Matrix yields and deforms plastically (at ε_{ym}) while the fibers continue to stretch elastically

The onset of composite failure begins as the fibers start to fracture, which corresponds to a strain of approximately ε_f^*

Continuous and Aligned Fiber Composites

- Composite failure is not catastrophic because
  - (1) not all fibers fracture at the same time as there will always be considerable variations in the fracture strength of brittle fiber materials
  - (2) even after fiber failure, the matrix is still intact inasmuch as $\varepsilon_f^* < \varepsilon_m^*$
- The fractured fibers, which are shorter than the original ones, are still embedded within the intact matrix and consequently are capable of sustaining a diminished load as the matrix continues to plastically deform

Continuous and Aligned Fiber Composites

❖ Elastic behaviour in Longitudinal loading

Assumption: Isostrain state

Fiber–matrix interfacial bond is very good, such that deformation of both matrix and fibers is the same

Then, the total load sustained by the composite F_c is equal to the sum of the loads carried by the matrix phase F_m and the fiber phase F_f ($F_c = F_m + F_f$)

Now $F = \sigma A$ (from the definition of stress) $\rightarrow \sigma_c A_c = \sigma_m A_m + \sigma_f A_f$ (where the subscripts c, m and f refer to composite, matrix and fiber, respectively)

Dividing by A_c we get $\sigma_c = \sigma_m A_m/A_c + \sigma_f A_f/A_c$

Since A_m/A_c and A_f/A_c are the area fractions of matrix and fiber, respectively, if the composite matrix and fiber phase lengths are equal, these area fractions are simply equivalent to the corresponding volume fractions

Writing the previous equation in terms of volume fractions we get
$\sigma_c = \sigma_m V_m + \sigma_f V_f$

Continuous and Aligned Fiber Composites

❖ Elastic behaviour in Longitudinal loading

The isostrain assumption implies $\varepsilon_c = \varepsilon_m = \varepsilon_f$ so we can write $\frac{\sigma_c}{\varepsilon_c} = \frac{\sigma_m}{\varepsilon_m} V_m + \frac{\sigma_f}{\varepsilon_f} V_f$

Finally, since we are discussing elastic deformation of composite (including fiber and matrix), σ/ε can be substituted by the corresponding Elastic modulus E to obtain

$$E_{cl} = E_m V_m + E_f V_f$$

This is the equation for modulus of elasticity of the continuous and aligned fiber composite in the longitudinal direction (hence the subscript 'cl' instead of just 'c')

Since $V_m + V_f = 1$, we can write $E_{cl} = E_m(1-V_f) + E_f V_f$

Similar relationship is true for density in terms of volume fraction of fibers

Also, the ratio of load carried by fibers to load carried by matrix for longitudinal loading is given by

$F_f/F_m = E_f V_f / E_m V_m$ (derive it yourself to check!)

Continuous and Aligned Fiber Composites

❖ Elastic behaviour in Transverse loading

This is an isostress state, i.e. $\sigma_c = \sigma_m = \sigma_f = \sigma$ and the strain is given by $\varepsilon_c = \varepsilon_m V_m + \varepsilon_f V_f$

Therefore, $\frac{\sigma}{E_{ct}} = \frac{\sigma}{E_m} V_m + \frac{\sigma}{E_f} V_f$ which gives $\frac{1}{E_{ct}} = \frac{V_m}{E_m} + \frac{V_f}{E_f}$ or

$E_{ct} = \frac{E_m E_f}{V_m E_f + V_f E_m} = \frac{E_m E_f}{(1-V_f) E_f + V_f E_m}$ (analogous to lower bound for particle reinforced composite, while the longitudinal loading case on previous slide is analogous to upper bound for particulate)

❖ Longitudinal Tensile strength

$\sigma_{cl}^* = \sigma_m' (1-V_f) + \sigma_f^* V_f$ (please refer to Figure on slide 6 in this set for σ_m' and σ_f^*)

❖ Transverse Tensile strength

Whereas longitudinal strength is dominated by fiber strength, a variety of factors will have a significant influence on the transverse strength; these factors include properties of both the fiber and matrix, the fiber–matrix bond strength, and the presence of voids

Measures that have been used to improve the transverse strength of these composites usually involve modifying properties of the matrix

Discontinuous and Aligned Fiber Composites

- Even though reinforcement efficiency is lower for discontinuous than for continuous fibers, discontinuous and aligned–fiber composites are becoming increasingly important in the commercial market. Chopped-glass fibers are used most extensively; however, carbon and aramid discontinuous fibers are also used. These short-fiber composites can be produced with moduli of elasticity and tensile strengths that approach 90% and 50%, respectively, of their continuous-fiber counterparts

- For a discontinuous and aligned–fiber composite having a uniform distribution of fibers and in which $l > l_c$, the longitudinal strength (σ_{cd}*) is given by the relationship

$$\sigma_{cd}^{*} = \sigma_f^{*} V_f \left(1 - \frac{l_c}{2l}\right) + \sigma_m'(1 - V_f)$$

where σ_f* and σ_m' represent, respectively, the fracture strength of the fiber and the stress in the matrix when the composite fails

- If the fiber length is less than critical ($l < l_c$) then the longitudinal strength ($\sigma_{cd'}$*) is

$$\sigma_{cd'}^{*} = \frac{l\tau_c}{d} V_f + \sigma_m'(1 - V_f)$$

where d is the fiber diameter and τ_c is the smaller of either the fiber–matrix bond strength or the matrix shear yield strength

Discontinuous and Randomly Oriented Fibers

- Normally, when the fiber orientation is random, short and discontinuous fibers are used. A similar *rule-of-mixtures* equation can be used for the elastic modulus

$$E_{cd} = KE_f V_f + E_m V_m$$

- In this expression, K is a fiber efficiency parameter that depends on V_f and the E_f/E_m ratio. Its magnitude will be less than unity, usually in the range 0.1 to 0.6. Thus, for random-fiber reinforcement (as with oriented-fiber reinforcement), the modulus increases with increasing volume fraction of fiber

Properties of Unreinforced and Reinforced Polycarbonates with Randomly Oriented Glass Fibers

		Value for Given Amount of Reinforcement (vol%)		
Property	***Unreinforced***	***20***	***30***	***40***
Specific gravity	1.19–1.22	1.35	1.43	1.52
Tensile strength [MPa (ksi)]	59–62 (8.5–9.0)	110 (16)	131 (19)	159 (23)
Modulus of elasticity [GPa (10^6 psi)]	2.24–2.345 (0.325–0.340)	5.93 (0.86)	8.62 (1.25)	11.6 (1.68)
Elongation (%)	90–115	4–6	3–5	3–5
Impact strength, notched Izod (lb_f/in.)	12–16	2.0	2.0	2.5

Effect of Fiber Orientation

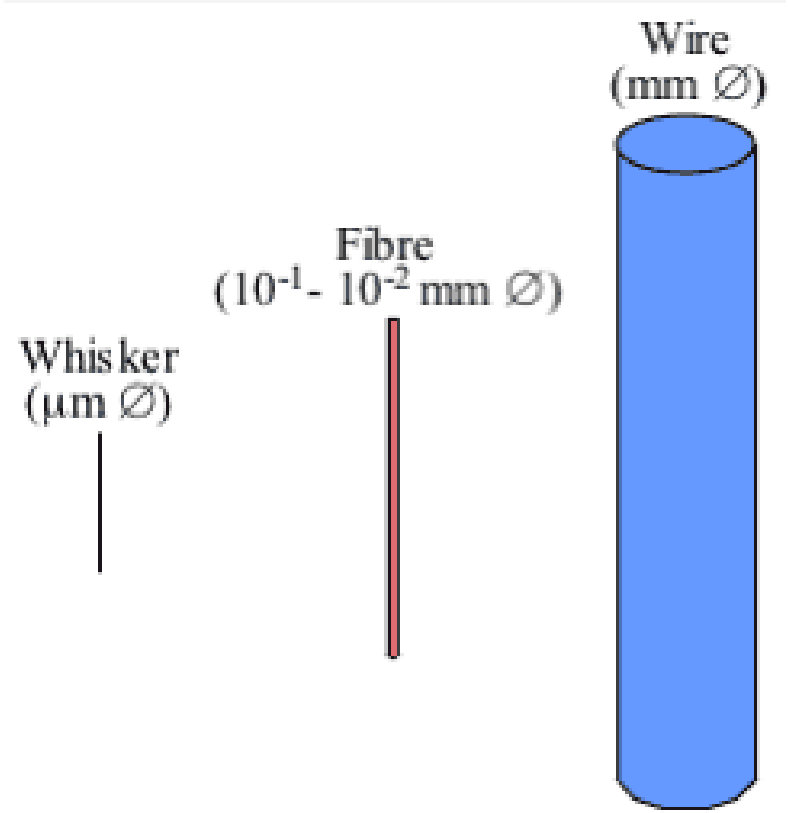
Fiber Orientation	*Stress Direction*	*Reinforcement Efficiency*
All fibers parallel	Parallel to fibers	1
	Perpendicular to fibers	0
Fibers randomly and uniformly distributed within a specific plane	Any direction in the plane of the fibers	$\frac{3}{8}$
Fibers randomly and uniformly distributed within three dimensions in space	Any direction	$\frac{1}{5}$

Reinforcement Efficiency of Fiber–Reinforced Composites for Several Fiber Orientations and at Various Directions of Stress Application

Types of Fibers

- On the basis of diameter and character, fibers are grouped into three different classifications: whiskers, fibers, and wires
- Whiskers are very thin single crystals that have extremely large length-to-diameter ratios
- High degree of crystalline perfection and are virtually flaw-free, which accounts for their exceptionally high strengths; they are among the strongest known materials
- But, extremely expensive and also, it is difficult and often impractical to incorporate whiskers into a matrix
- Whisker materials include graphite, silicon carbide, silicon nitride, and aluminum oxide
- Materials that are classified as fibers are either polycrystalline or amorphous and have small diameters
- Fibrous materials are generally either polymers or ceramics (e.g., the polymer aramids, glass, carbon, boron, aluminum oxide, and silicon carbide)
- Wires have relatively large diameters
- Typical materials include steel, molybdenum, and tungsten (e.g. radial steel reinforcement in automobile tires, in filament-wound rocket casings, and in wire-wound high-pressure hoses)

Types of Fibers



Functions of the Matrix Phase

- For fiber-reinforced composites, the matrix phase serves several functions
  - Binds the fibers together and acts as the medium by which an externally applied stress is transmitted and distributed to the fibers
  - Protects the individual fibers from surface damage as a result of mechanical abrasion or chemical reactions with the environment (such interactions may introduce surface flaws capable of forming cracks, which may lead to failure at low tensile stress levels)
  - Finally, the matrix separates the fibers and, by virtue of its relative softness and plasticity, prevents the propagation of brittle cracks from fiber to fiber, which could result in catastrophic failure
- Also, it is essential that adhesive bonding forces between fiber and matrix be high to minimize fiber pullout
- Adequate bonding is essential to maximize the stress transmittance from the weak matrix to the strong fibers

Polymer Matrix Composites

- Polymer-matrix composites (PMCs) consist of a polymer resin as the matrix and fibers as the reinforcement medium. These materials are used in the greatest diversity of composite applications, as well as in the largest quantities, in light of their room-temperature properties, ease of fabrication, and cost

- Note: The term resin is used in this context to denote a high-molecular-weight reinforcing plastic

- Various classifications of PMCs are done according to reinforcement type i.e., glass, carbon, and aramid

Glass Fiber-Reinforced Polymer Composites

- **Fiberglass** is simply a composite consisting of glass fibers, either continuous or discontinuous, contained within a polymer matrix; this type of composite is produced in the largest quantities

- The glass that is most commonly drawn into fibers is sometimes referred to as E-glass. Fiber diameters normally range between 3 and 20 μm.

- **Advantages** of Glass as a fiber reinforcement material

  (1) Easily drawn into high-strength fibers from the molten state
  (2) Readily available and may be fabricated into a glass-reinforced plastic economically
  (3) Relatively strong fiber that can produce very high specific strength composites
  (4) Possesses a chemical inertness making the composite useful in corrosive environments

- **Limitations:**
  (1) Not very stiff and do not display the rigidity that is necessary for some applications (e.g., as structural members for airplanes and bridges)
  (2) Limited to service temperatures below 200°C; at higher temperatures, most polymers begin to flow or to deteriorate (high-purity fused silica fibers and high T polymers such as the polyimide resins → up to 300°C

- **Applications:** automotive and marine bodies, plastic pipes, storage containers, and industrial floorings

  - The transportation industries are using increasing amounts of glass fiber–reinforced plastics in an effort to decrease vehicle weight and boost fuel efficiencies

Carbon Fiber-Reinforced Polymer Composites

- **Carbon** is a high-performance fiber material that is the most commonly used reinforcement in advanced (i.e., nonfiberglass) polymer-matrix composites.

- **Advantages** of Carbon as a reinforcement material

(1) Carbon fibers have high specific moduli and specific strengths
(2) They retain their high tensile modulus and high strength at elevated temperatures; high-temperature oxidation, however, may be a problem
(3) At room temperature, carbon fibers are not affected by moisture or a wide variety of solvents/acids/bases
(4) These fibers exhibit a diversity of physical and mechanical characteristics, allowing composites incorporating these fibers to have specific engineered properties
(5) Relatively inexpensive/cost-effective fiber- and composite-manufacturing processes

- **Classification** based on tensile modulus: standard, intermediate, high, and ultrahigh moduli.

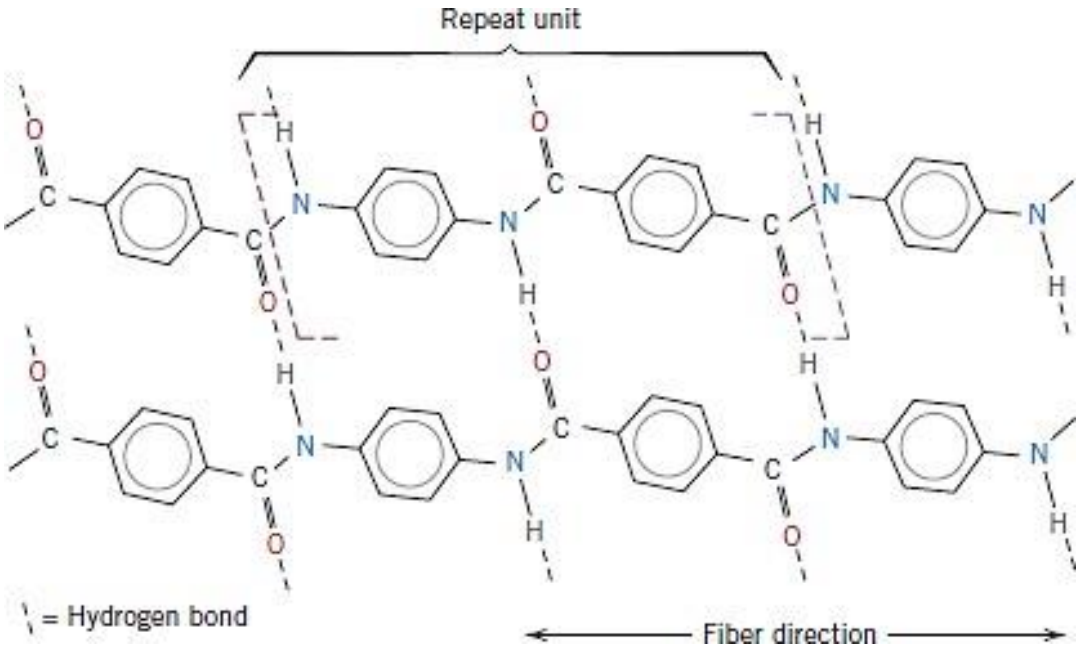
- Fiber diameters normally range between 4 and 10 μm; both continuous and chopped forms are available

- In addition, carbon fibers are normally coated with a protective epoxy size that also improves adhesion with the polymer matrix

- **Applications:** Used extensively in sports and recreational equipment (fishing rods, golf clubs), filament-wound rocket motor cases, pressure vessels, and aircraft structural components—military and commercial, both fixed-wing aircraft and helicopters (e.g., as wing, body, stabilizer, and rudder components)

Aramid Fiber-Reinforced Polymer Composites

- **Aramid** fibers are high-strength, high-modulus materials that were introduced in the early 1970s
- Outstanding strength-to-weight ratios (superior than metals)
- Chemically, this group of materials is known as poly(paraphenylene terephthalamide)
- Trade names for two of the most common are Kevlar and Nomex
- There are several grades (Kevlar 29, 49, and 149) that have different mechanical behaviors
- During synthesis, the rigid molecules are aligned in the direction of the fiber axis, as liquid crystal domains; the repeat unit and the mode of chain alignment are represented below



Aramid Fiber-Reinforced Polymer Composites

- Mechanically, these fibers have longitudinal tensile strengths and tensile moduli that are higher than those of other polymeric fiber materials; however, they are relatively weak in compression

- In addition, this material is known for its toughness, impact resistance, and resistance to creep and fatigue failure. Despite being thermoplastics, they are resistant to combustion and stable to relatively high T (retain their high mechanical properties between –200°C and 200°C)

- Chemically, they are susceptible to degradation by strong acids and bases, but they are relatively inert in other solvents and chemicals

- Aramid fibers are most often used in composites having polymer matrices; common matrix materials are the epoxies and polyesters

- Applications: Ballistic products (bulletproof vests and armour), sporting goods, tires, ropes, missile cases, and pressure vessels and as a replacement for asbestos in automotive brake and clutch linings and gaskets

Metal-Matrix Composites (MMC)

- Matrix phase is a ductile metal and reinforcement is done to be able to use the composite at higher service temperatures than the base metal or other properties such as higher specific stiffness, specific strength, abrasion resistance, creep resistance, thermal conductivity, and dimensional stability

- The advantages of MMCs over Polymer-matrix composites (PMCs) include higher operating temperatures, nonflammability and greater resistance to degradation by organic fluids while the disadvantage is that MMCs are more expensive than PMCs

- Usual matrix materials include alloys of Al/Mg/Ti/Cu or superalloys; reinforcement may be in the form of particulates, both continuous and discontinuous fibers, and whiskers; concentrations normally range between 10 and 60 vol%. Continuous-fiber materials include carbon, silicon carbide, boron, aluminum oxide, and the refractory metals. However, discontinuous reinforcements consist primarily of silicon carbide whiskers, chopped fibers of aluminum oxide and carbon, and particulates of silicon carbide and aluminum oxide

- Sometimes, protective surface coating to the reinforcement or tuning the matrix alloy composition is necessary to prevent high temperature degradation

- Processing of MMCs – 2 steps – (1) Consolidation/Synthesis (2) Shaping operation

Metal-Matrix Composites (MMC)

- Consolidation/Synthesis refers to the introduction of the reinforcement in the matrix, while the subsequent shaping operation can be done by standard metal forming operations like rolling, forging or extrusion in case of discontinuous-fiber MMCs

- **Applications**
  - In automobile industry as engine components made of Al-alloy matrix reinforced with Al-oxide and C fibers (light weight, wear resistance and thermal distortion resistance)
  - Also used in driveshafts, extruded stabilizer bars, forged suspension and transmission components
  - Advanced Al-alloy MMCs in aerospace industry due to their low density and control over thermal and mechanical properties
  - Continuous graphite fibers used as reinforcement for an antenna boom on the Hubble Space Telescope
  - GPS satellites use SiC-Al and graphite–Al MMCs for electronic packaging and thermal management systems

- The high-temperature creep and rupture properties of some superalloys (Ni- and Co-based alloys) may be enhanced by fiber reinforcement using refractory metals such as tungsten (application in turbine engines)

Ceramic-Matrix Composites (CMC)

- Fracture toughness of ceramics improved significantly by development of CMCs, which consist of particles, fibers or whiskers of one ceramic material embedded in a matrix of another ceramic
- The improvement in fracture toughness results from interactions between advancing cracks and dispersed phase particles – cracks initiate in the matrix but crack propagation/growth is hindered by the particles/fibers/whiskers
- Crack propagation may be retarded by various toughening mechanisms such as transformation toughening (described in detail in the next slide), or use of ceramic whiskers (e.g. SiC/Si_3N_4); These whiskers may inhibit crack propagation by (1) deflecting crack tips, (2) forming bridges across crack faces, (3) absorbing energy during pullout as the whiskers debond from the matrix, and/or (4) causing a redistribution of stresses in regions adjacent to the crack tips
- In general, increasing fiber content improves strength and fracture toughness
- CMCs exhibit improved high-temperature creep behavior and resistance to thermal shock
- Ceramic-matrix composites may be fabricated using hot pressing, hot isostatic pressing, and liquid-phase sintering techniques
- **Applications** SiC whisker-reinforced alumina is used as cutting-tool inserts for machining hard metal alloys; tool lives for these materials are greater than for cemented carbides

Ceramic-Matrix Composites (CMC)

Transformation toughening uses a phase transformation to arrest the propagation of cracks

Small particles of partially stabilized zirconia are dispersed within the matrix material, often Al_2O_3 or ZrO_2 itself

Typically, CaO, MgO, Y_2O_3, and CeO are used as stabilizers

Partial stabilization → metastable tetragonal phase rather than the stable monoclinic phase

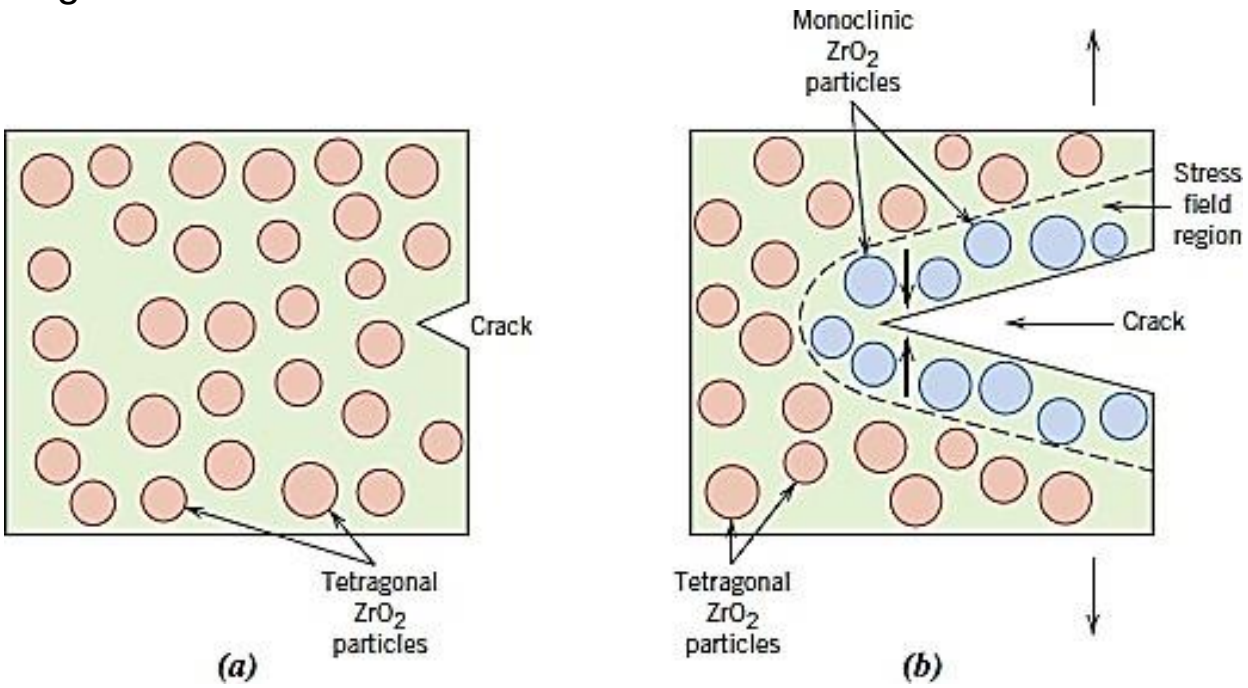
Stress field in front of a propagating crack causes these metastably retained tetragonal particles to undergo transformation to the stable monoclinic phase

Accompanying this transformation is a slight particle volume increase, and the net result is that compressive stresses are established on the crack surfaces near the crack tip that tend to pinch the crack shut, thereby arresting its growth

Transformation Toughening Mechanism

(a) A crack prior to inducement of the ZrO_2 particle phase transformation

(b) Crack arrestment due to the stress-induced phase transformation



Carbon-Carbon Composites

- Carbon-carbon composites refer to carbon fiber–reinforced carbon-matrix composite
- These materials are relatively new and expensive (due to complex processing techniques)
- High tensile moduli and tensile strengths, which are retained to T>2000°C
- Creep resistance and relatively large fracture toughness values
- Low coefficients of thermal expansion and relatively high thermal conductivities
- Hence, relatively low susceptibility to thermal shock
- Major drawback is a propensity to high-temperature oxidation
- **Applications** in rocket motors, as friction materials in aircraft and high-performance automobiles, for hot-pressing molds, in components for advanced turbine engines, and as ablative shields for re-entry vehicles

Hybrid Composites

- Two or more different kinds of fibers in a single matrix → better all-around combination of properties than composites containing only a single fiber type
- Most common system → both carbon and glass fibers are incorporated into a polymeric resin
- Carbon fibers are strong and relatively stiff and provide a low-density reinforcement; however, they are expensive. Glass fibers are inexpensive and lack the stiffness of carbon. The glass–carbon hybrid is stronger and tougher, has a higher impact resistance, and may be produced at a lower cost than either of the comparable all-carbon or all-glass reinforced plastics
- The two different fibers may be combined in a number of ways, which will ultimately affect the overall properties. For example, the fibers may all be aligned and intimately mixed with one another, or laminations may be constructed consisting of layers, each of which consists of a single fiber type, alternating with one another. In virtually all hybrids, the properties are anisotropic
- When hybrid composites are stressed in tension, failure is usually noncatastrophic (i.e., does not occur suddenly). The carbon fibers are the first to fail, at which time the load is transferred to the glass fibers. Upon failure of the glass fibers, the matrix phase must sustain the applied load. Eventual composite failure concurs with that of the matrix phase
- Principal applications for hybrid composites are lightweight land, water, and air transport structural components, sporting goods, and lightweight orthopaedic components

Processing of Fiber-Reinforced Composites

Pultrusion is used for the manufacture of components having continuous lengths and a constant cross-sectional shape (rods, tubes, beams, etc.)

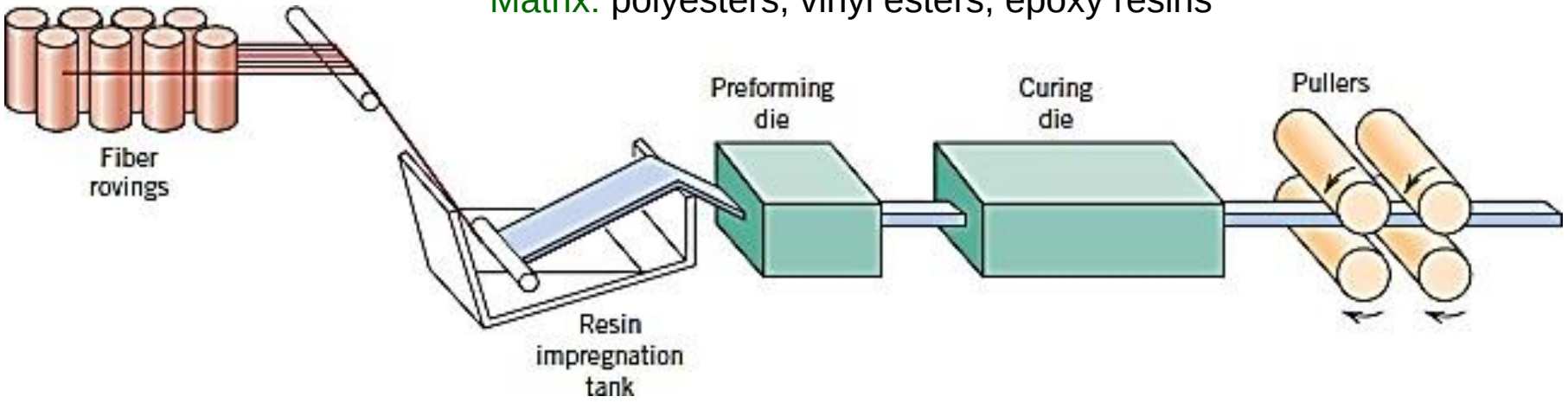
Continuous-fiber rovings or tows are first impregnated with a thermosetting resin and then pulled through a steel die that preforms to the desired shape and also establishes the resin/fiber ratio

The stock then passes through a curing die that is precision machined so as to impart the final shape; this die is also heated to initiate curing of the resin matrix

A pulling device draws the stock through the dies and also determines the production speed

Tubes and hollow sections are made possible by using center mandrels or inserted hollow cores.

Reinforcement: Glass, carbon, aramid fibers (40-70 vol. %)
Matrix: polyesters, vinyl esters, epoxy resins



Processing of Fiber-Reinforced Composites

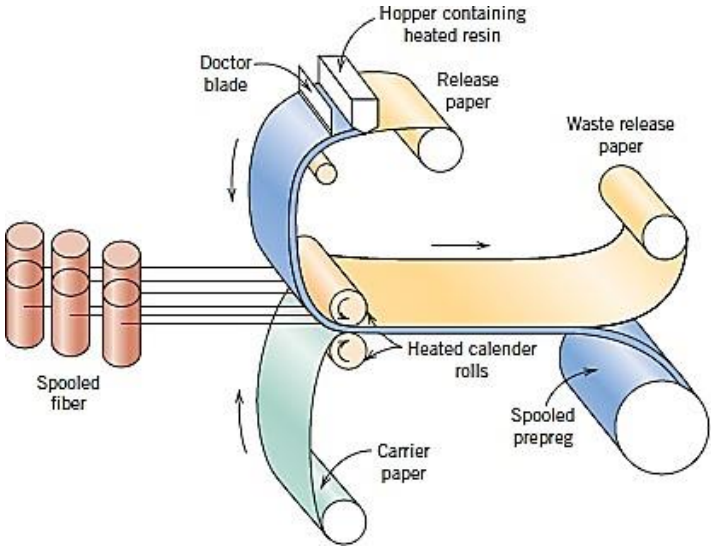
Prepreg is the composite industry's term for continuous-fiber reinforcement preimpregnated with a polymer resin that is only partially cured

First, collimate a series of spool-wound continuous-fiber tows and then sandwich and press them between sheets of release and carrier paper using heated rollers, a process termed calendering

The release paper sheet has been coated with a thin film of heated resin solution of relatively low viscosity so as to provide for its thorough impregnation of the fibers

A doctor blade spreads the resin into a film of uniform thickness and width

The final prepreg product—the thin tape consisting of continuous and aligned fibers embedded in a partially cured resin—is prepared for packaging by winding onto a cardboard core.



Typical tape thicknesses range between 0.08 and 0.25 mm; and tape widths range between 25 and 1525 mm (1 and 60 in.)

Resin content 35 to 45 vol.%.

Both thermoplastic and thermosetting resins are used; carbon, glass, and aramid fibers are the common reinforcements.

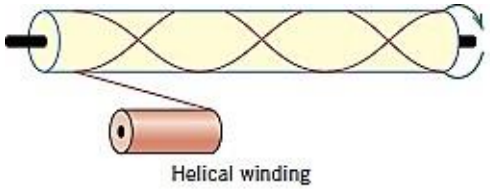
Processing of Fiber-Reinforced Composites

Filament winding Continuous reinforcing fibers are accurately positioned in a predetermined pattern to form a hollow (usually cylindrical) shape

The fibers, either as individual strands or as tows, are first fed through a resin bath and then are continuously wound onto a mandrel, usually using automated winding equipment

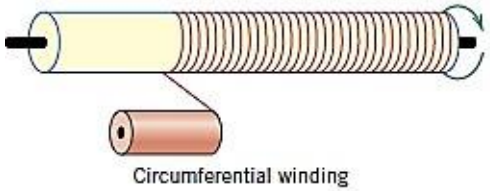
After the appropriate number of layers have been applied, curing is carried out either in an oven or at room temperature, after which the mandrel is removed

Various winding patterns are possible (i.e., circumferential, helical, and polar) to give the desired mechanical characteristics



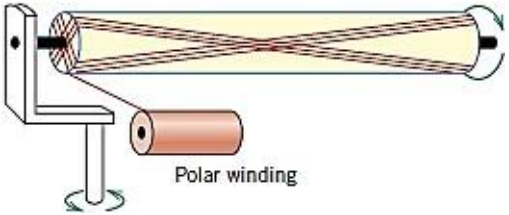
Filament-wound parts have very high strength-to-weight ratios

Also, a high degree of control over winding uniformity and orientation



When automated, the process is most economically attractive

Common filament-wound structures include rocket motor casings, storage tanks and pipes, and pressure vessels.



Structural Composites

- A structural composite is a multi-layered and normally low-density composite used in applications requiring structural integrity, ordinarily high tensile, compressive, and torsional strengths and stiffnesses

- The properties of these composites depend not only on the properties of the constituent materials, but also on the geometrical design of the structural elements

- Laminar composites and sandwich panels are two of the most common structural composites

- A laminar composite is composed of two-dimensional sheets or panels (plies or laminae) bonded to one another. Each ply has a preferred high-strength direction, such as is found in continuous and aligned fiber–reinforced polymers. A multi-layered structure such as this is termed a laminate

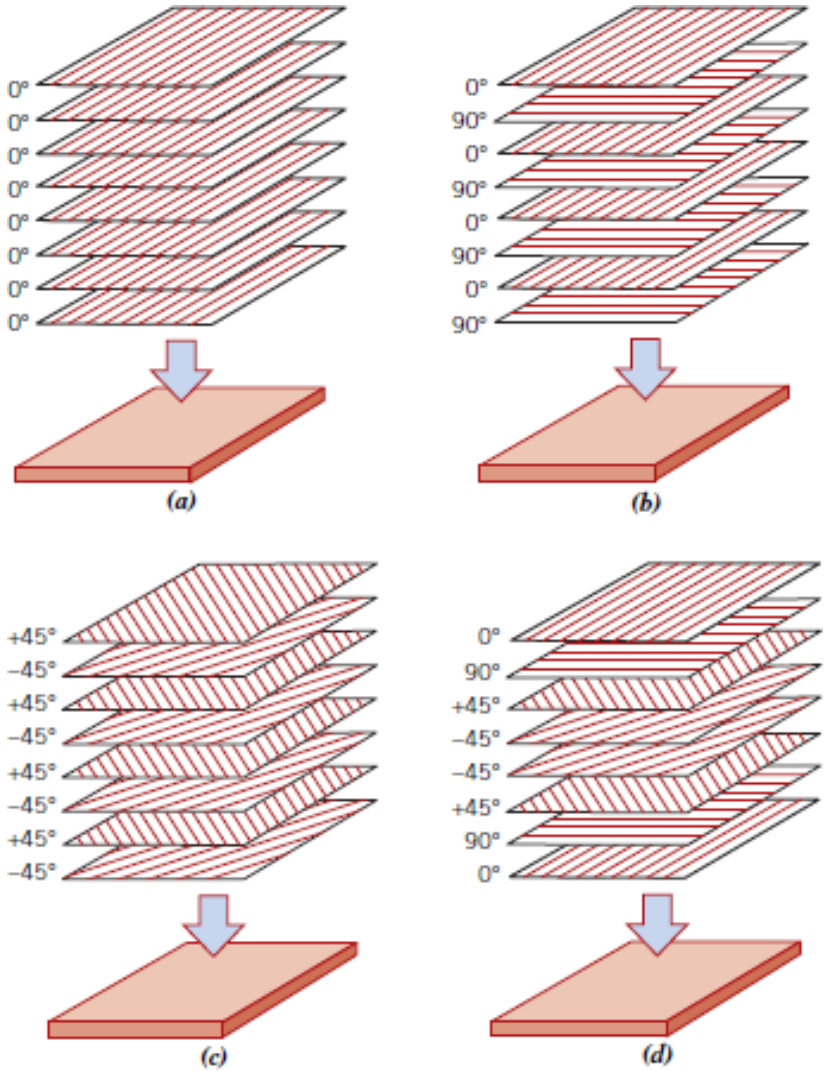
- Laminate properties depend on several factors, including how the high-strength direction varies from layer to layer. In this regard, there are four classes of laminar composites: unidirectional, cross-ply, angle-ply, and multidirectional

- **Applications** that use laminate composites are primarily in aircraft, automotive, marine, and building/civil-infrastructure sectors. Specific applications include: aircraft—fuselage, vertical and horizontal stabilizers, landing-gear hatch, floors, fairings, and rotor blades for helicopters; automotive—automobile panels, sports car bodies, and drive shafts; marine—ship hulls, hatch covers, deckhouses, bulkheads, and propellers; building/civil-infrastructure—bridge components, long-span roof structures, beams, structural panels, roof panels, and tanks. Laminates are also used extensively in sports and recreation equipment, e.g. the modern ski

Structural Composites

- For unidirectional, the orientation of the high-strength direction for all laminae is the same
- Cross-ply laminates have alternating high-strength layer orientations of 0° and 90°
- For angle-ply, successive layers alternate between +θ and – θ high-strength orientations (e.g., ±45°)
- The multidirectional laminates have several high-strength orientations
- For virtually all laminates, layers are typically stacked such that fiber orientations are symmetric relative to the laminate midplane; this arrangement prevents any out-of-plane twisting or bending
- In-plane properties (e.g., modulus of elasticity and strength) of unidirectional laminates are highly anisotropic. Cross-, angle-, and multidirectional laminates are designed to increase the degree of in-plane isotropy; multidirectional can be fabricated to be most isotropic; degree of isotropy decreases with angle- and cross-ply materials

Various laminar composites
(a) Unidirectional (b) Cross-ply
(c) Angle-ply (d) Multidirectional



Structural Composites

- Sandwich panels are designed to be lightweight beams or panels having relatively high stiffnesses and strengths. A sandwich panel consists of two outer sheets, faces, or skins that are separated by and adhesively bonded to a thicker core

- The outer sheets are made of a relatively stiff and strong material, typically aluminum alloys, steel and stainless steel, fiber-reinforced plastics, and plywood; they carry bending loads that are applied to the panel. When a sandwich panel is bent, one face experiences compressive stresses, the other tensile stresses

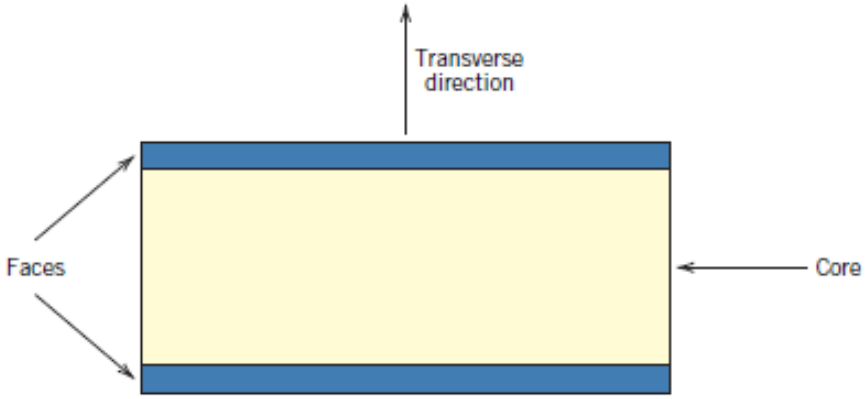
- The core material is lightweight and normally has a low modulus of elasticity. The core provides continuous support for the faces and holds them together. In addition, it must have sufficient shear strength to withstand transverse shear stresses and also be thick enough to provide high shear stiffness (to resist buckling of the panel)

- Tensile and compressive stresses on the core are much lower than on the faces. Panel stiffness depends primarily on the properties of the core material and core thickness; bending stiffness increases significantly with increasing core thickness. Furthermore, it is essential that faces be bonded strongly to the core

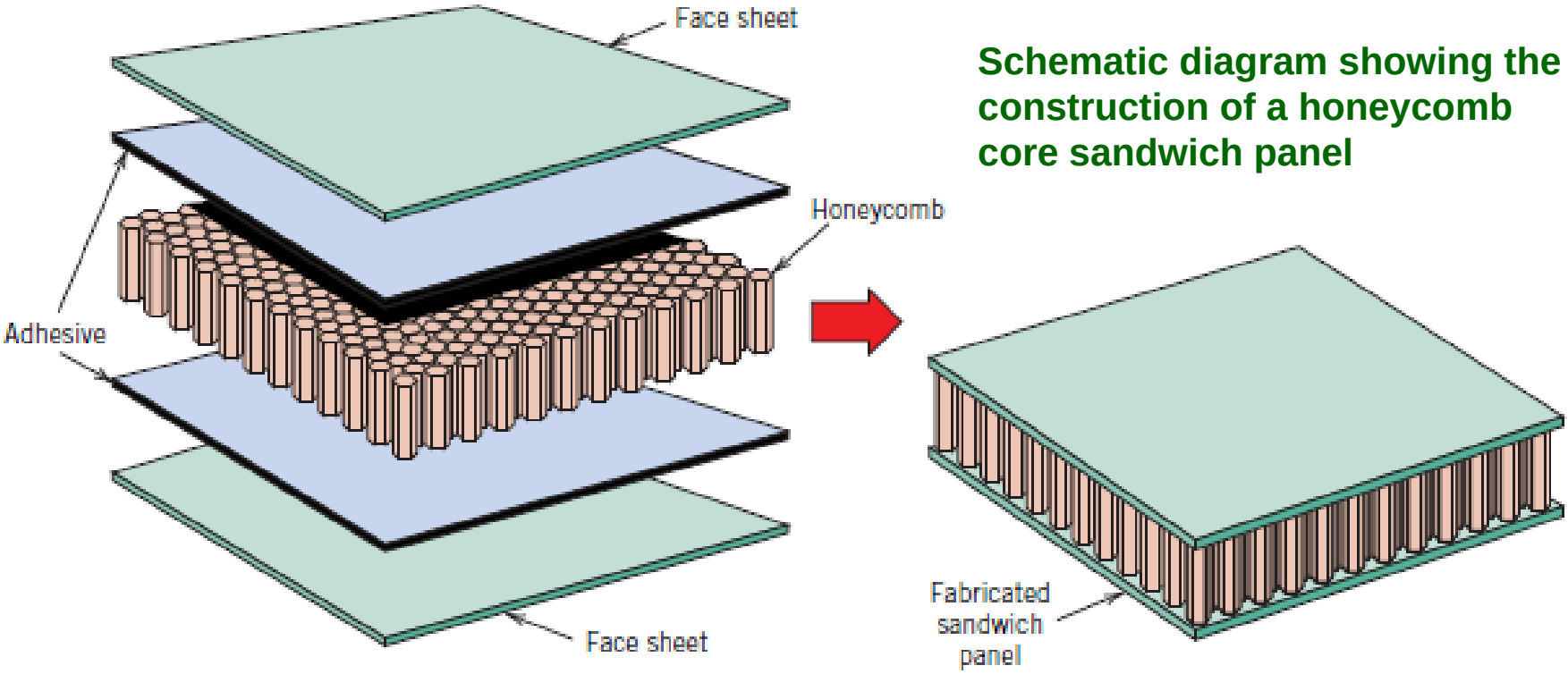
- Sandwich panel is a cost-effective composite because core materials are less expensive than materials used for the faces

- Core materials typically fall within three categories: rigid polymeric foams, wood, and honeycombs

Structural Composites



Schematic diagram showing the cross-section of a sandwich panel



Schematic diagram showing the construction of a honeycomb core sandwich panel

Nanocomposites

- Nanocomposites are composed of nanosized particles (or nanoparticles) that are embedded in a matrix material
- Physical and chemical properties experience dramatic changes with decrease in size of the nanoparticles and the degree of change depends on particle size (i.e., number of atoms)
- Two factors account for the size-induced properties of nanoparticles: (1) the increase in ratio of particle surface area to volume and (2) particle size
- Surface atoms behave differently than atoms located in the interior of a material. Consequently, as the size of a particle decreases, the relative ratio of surface atoms to bulk atoms increases; this means that surface phenomena begin to dominate. Furthermore, for extremely small particles, quantum effects begin to appear
- Although nanocomposite matrix materials may be metals and ceramics, the most common matrices are polymers
- The properties of a nanocomposite depend not only on the properties of both matrix and nanoparticle, but also on nanoparticle shape and content as well as matrix–nanoparticle interfacial characteristics
- Most of today's commercial nanocomposites use three general nanoparticle types: nanocarbons, nanoclays, and particulate nanocrystals

Nanocomposites

- Nanoparticle loadings (i.e., contents) vary significantly and depend on application. For example, carbon nanotube concentrations on the order of 5 wt.% can lead to significant increases in strength and stiffness. However, between 15 and 20 wt.% of carbon nanotubes are required to produce electrical conductivities necessary for some applications (e.g., to protect a nanocomposite structure from experiencing electrostatic discharges)

- One of the main challenges in the production of nanocomposite materials is processing. For most applications, the nanosize particles must be dispersed uniformly and homogeneously within the matrix

- Some examples of nanocomposite materials in various industries are as follows

  - Nanocomposite thin film bags/containers for food/beverage packaging
  - Nanocomposite coatings for automobile tires (to increase air pressure retention)
  - Graphene nanocomposites in anodes for lithium-ion rechargeable batteries
  - Flame barrier coatings made of multi-walled carbon nanotubes dispersed in silicone matrix
  - Polymer nanocomposites in dental restoration (filling) materials
  - Multi-walled C nanotubes in epoxy resins for wind turbine blades/sports equipment
  - Electrostatic dissipation in automotive and aircraft polymer fuel lines - the motion of highly flammable fuels can lead to the production of static charges, with risk of spark generation and the possibility of explosion → However, dissipation of such charge buildups can occur if the fuel lines are made electrically conductive by incorporating multi-walled carbon nanotubes into the polymer

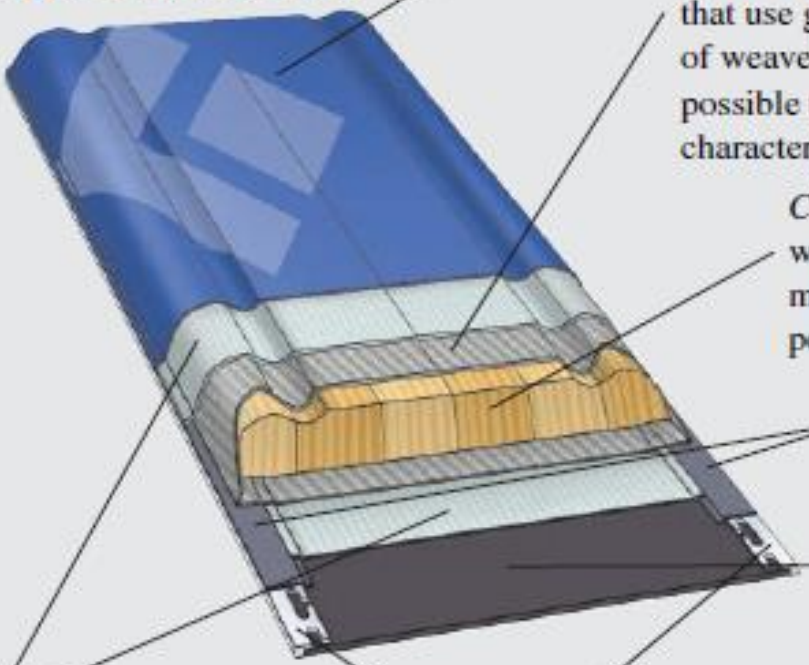
Composites in Modern Ski

Top sheet. Polyamide polymer that has a relatively low glass transition temperature and resists chipping.

Courtesy of Black Diamond Equipment, Ltd.

Torsion box wrap. Fiber-reinforced composites that use glass, aramid, or carbon fibers. A variety of weaves and weights of reinforcement are possible that are utilized to "tune" the flexural characteristics of the ski.

Core. Foam, vertical laminates of wood, wood-foam laminates, honeycomb, and other materials. Commonly used woods include poplar, spruce, bamboo, balsa, and birch.



Vibration-absorbing material. Rubber is normally used.

Base. Ultra-high-molecular-weight polyethylene is used because of its low coefficient of friction and abrasion resistance.

Reinforcement layers. Fiber-reinforced composites that normally use glass fibers. A variety of weaves and weights of reinforcement are possible to provide longitudinal stiffness.

Edges. Carbon steel that has been treated to have a hardness of 48 HRC. Facilitates turning by "cutting" into the snow.

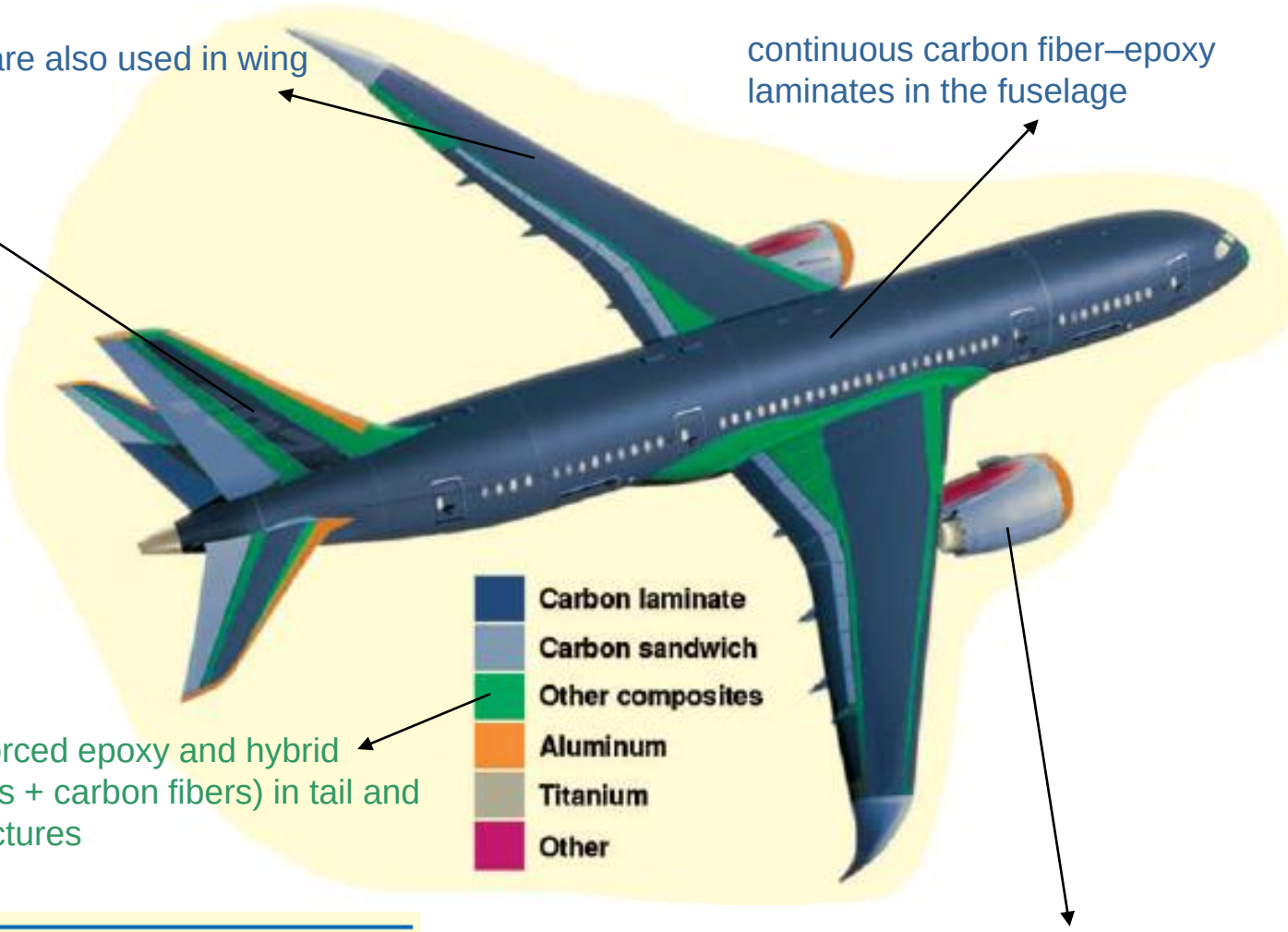
(a)



Composites in Boeing 787 Dreamliner

carbon laminates are also used in wing and tail structures

continuous carbon fiber–epoxy laminates in the fuselage

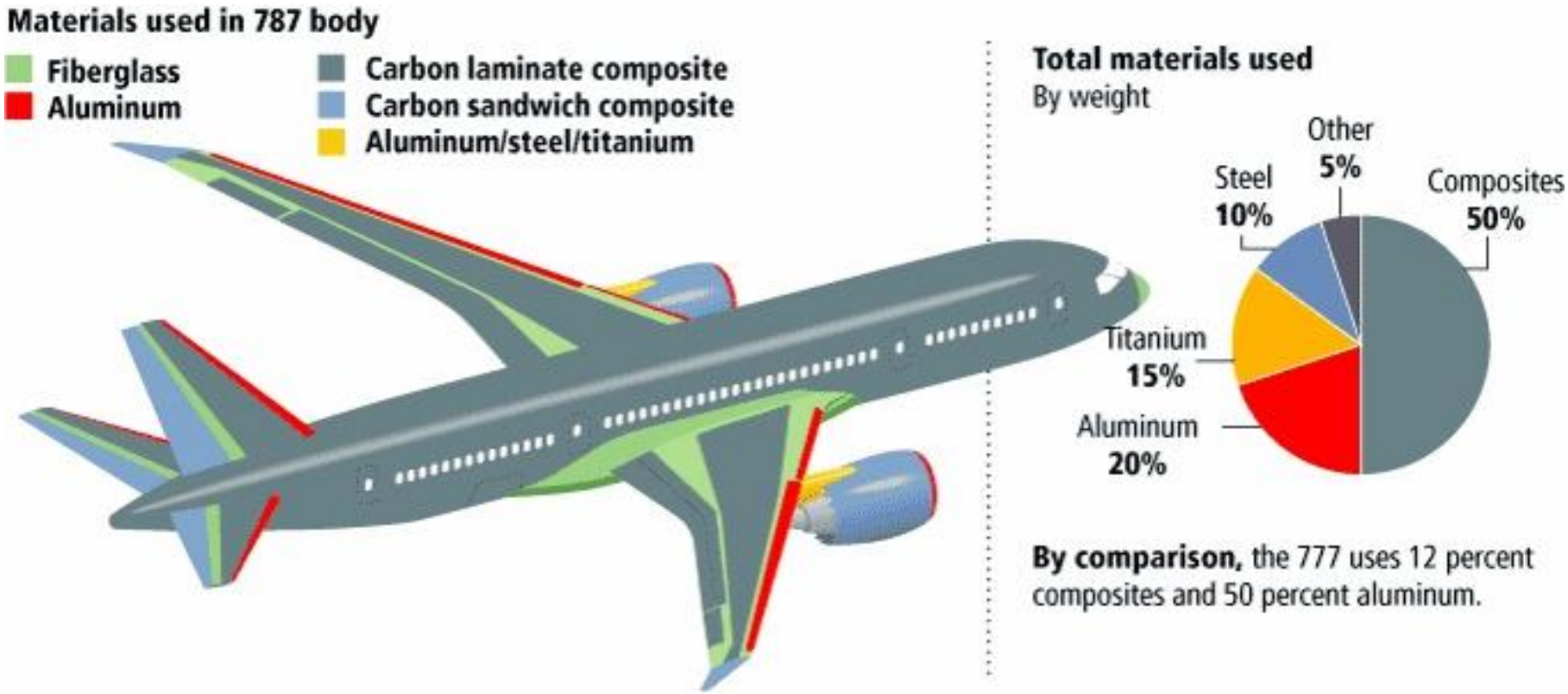


glass fiber–reinforced epoxy and hybrid composites (glass + carbon fibers) in tail and trailing wing structures

Aircraft	Material Content (Weight Percent)				
	Composites	Al Alloys	Ti Alloys	Steel	Other
787	50	20	15	10	5
777	11	70	7	11	1

Sandwich panels in nacelles (i.e., housing structures surrounding the engines) as well as trailing tail components; faces made of carbon fiber-epoxy laminates and cores are honeycomb structures of Al alloy sheet

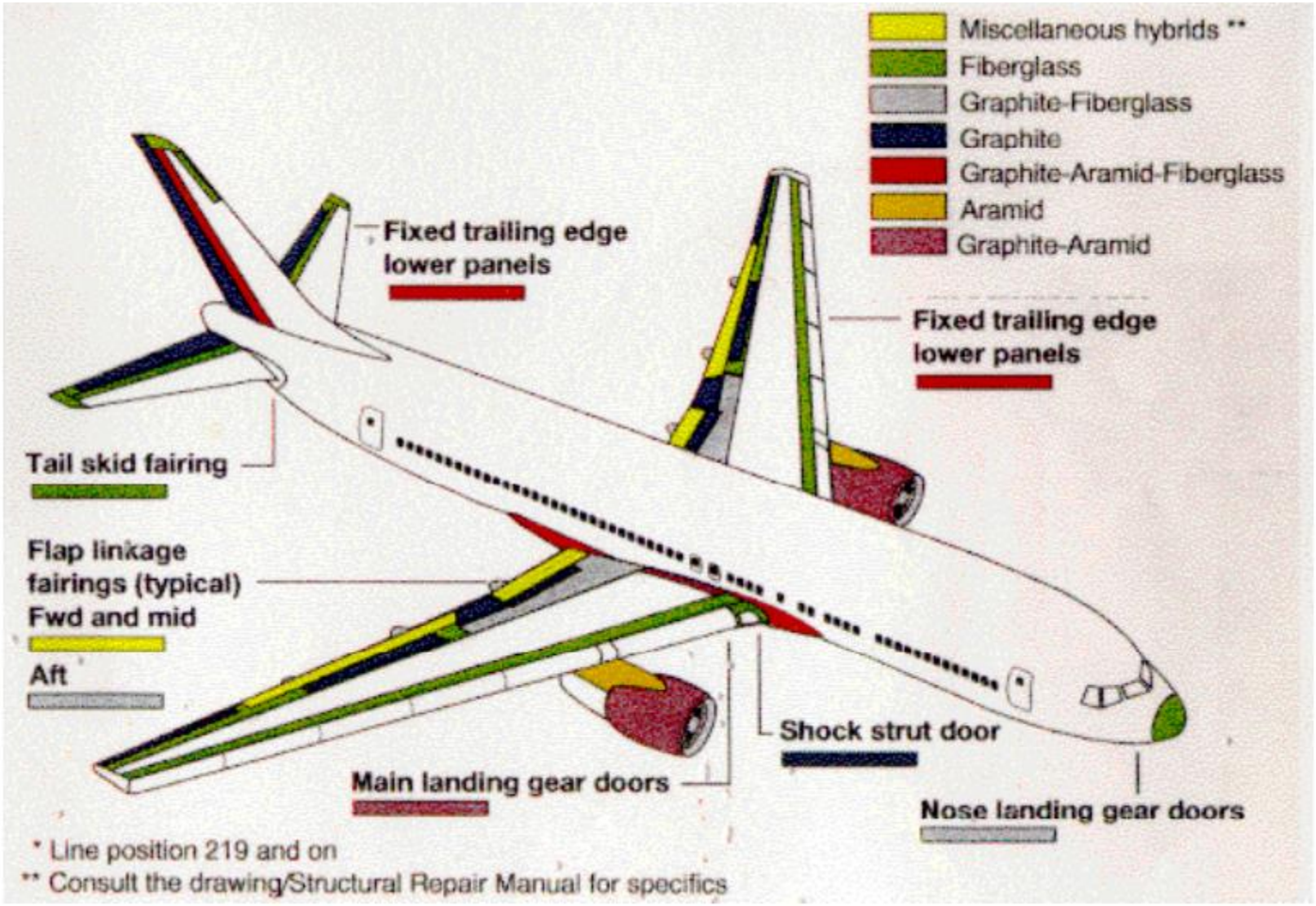
Some more on composites in planes/cars, etc.



https://www.appropedia.org/Composites_in_the_Aircraft_Industry

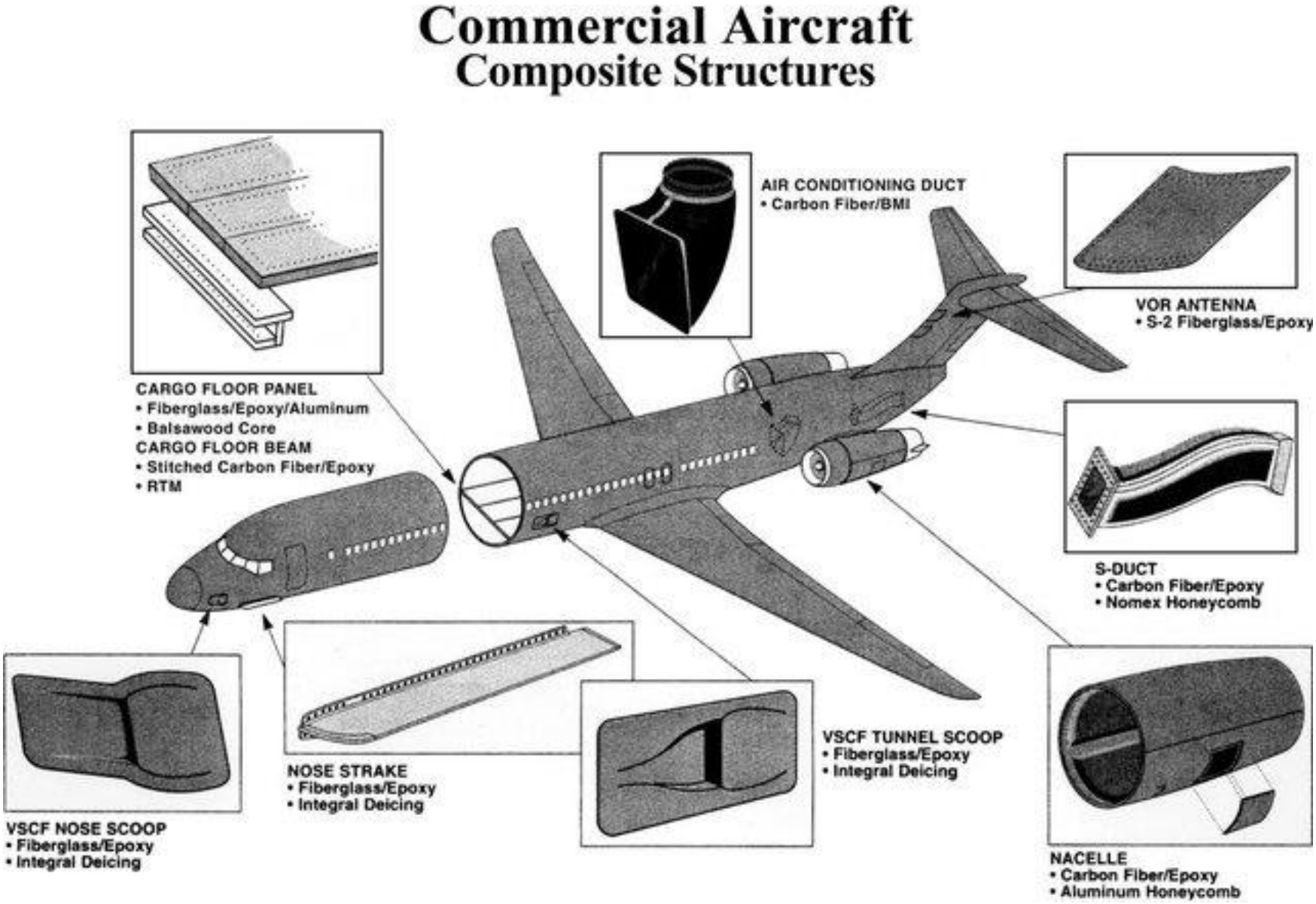
Some more on composites in planes/cars, etc.

Composites Parts Used in Boeing 767



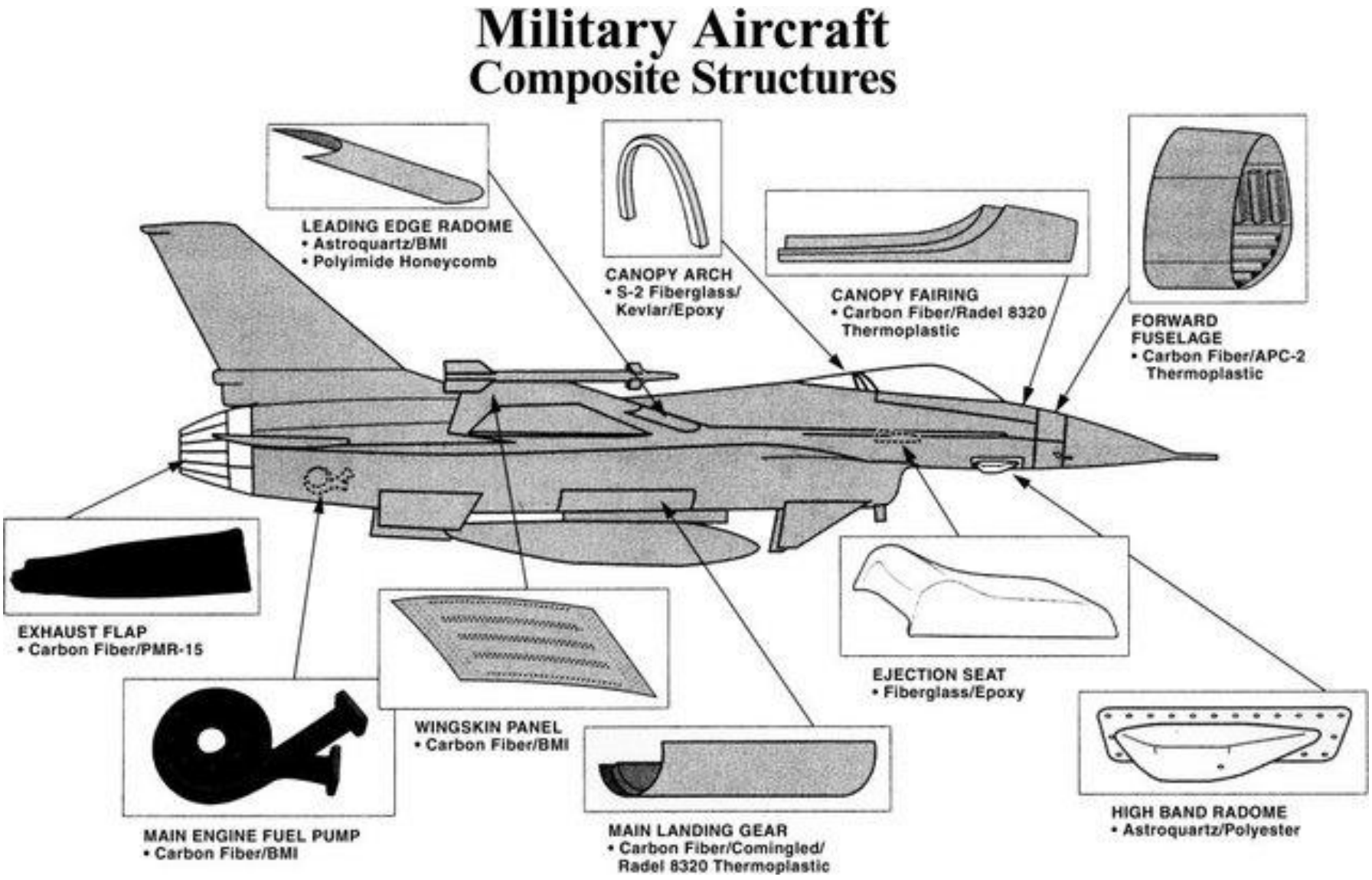
https://advancedmaterials2018.wordpress.com/2018/07/02/some-interesting-facts-about-air-craft-composite-materials/

Some more on composites in planes/cars, etc.



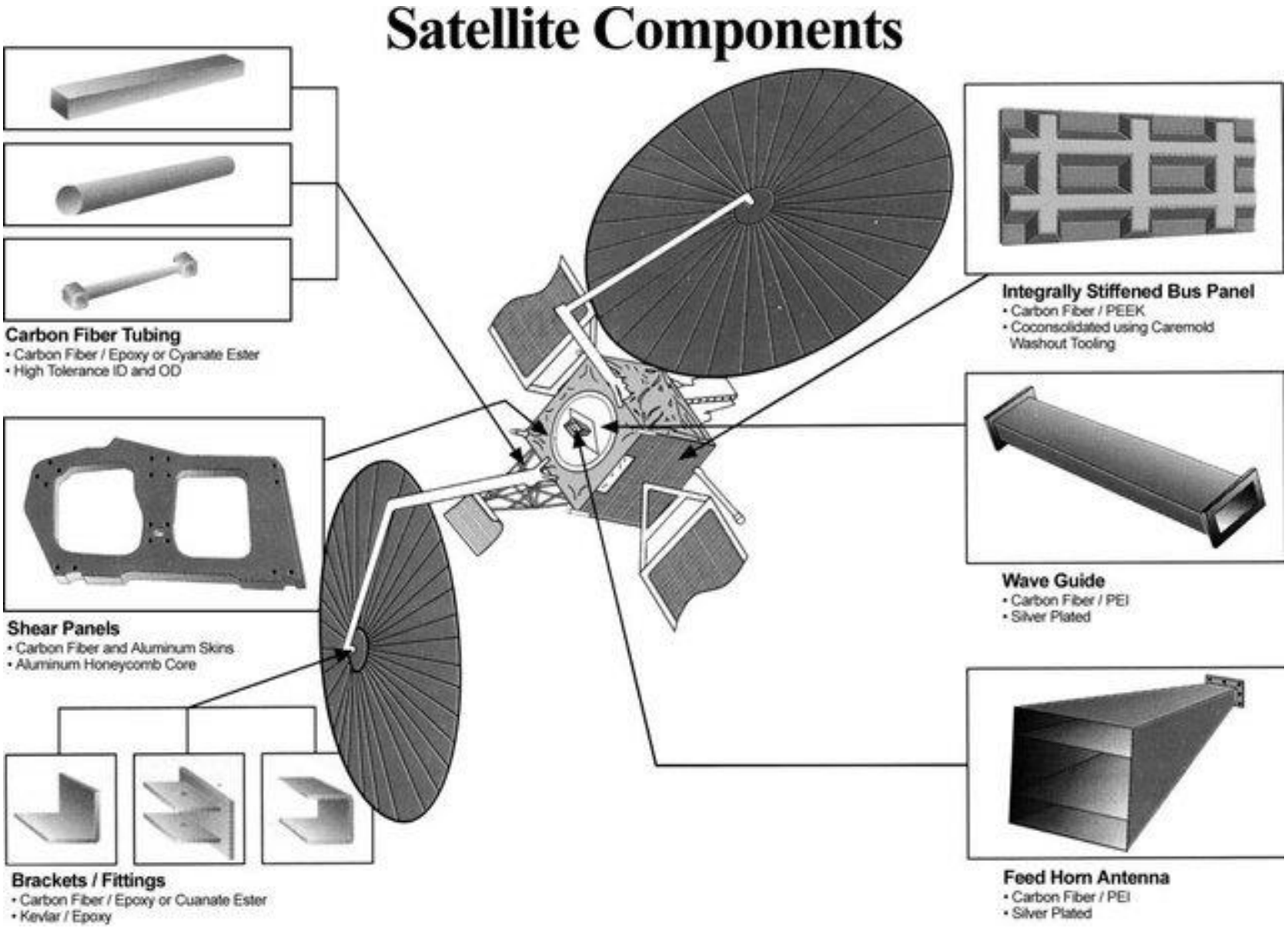
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Some more on composites in planes/cars, etc.



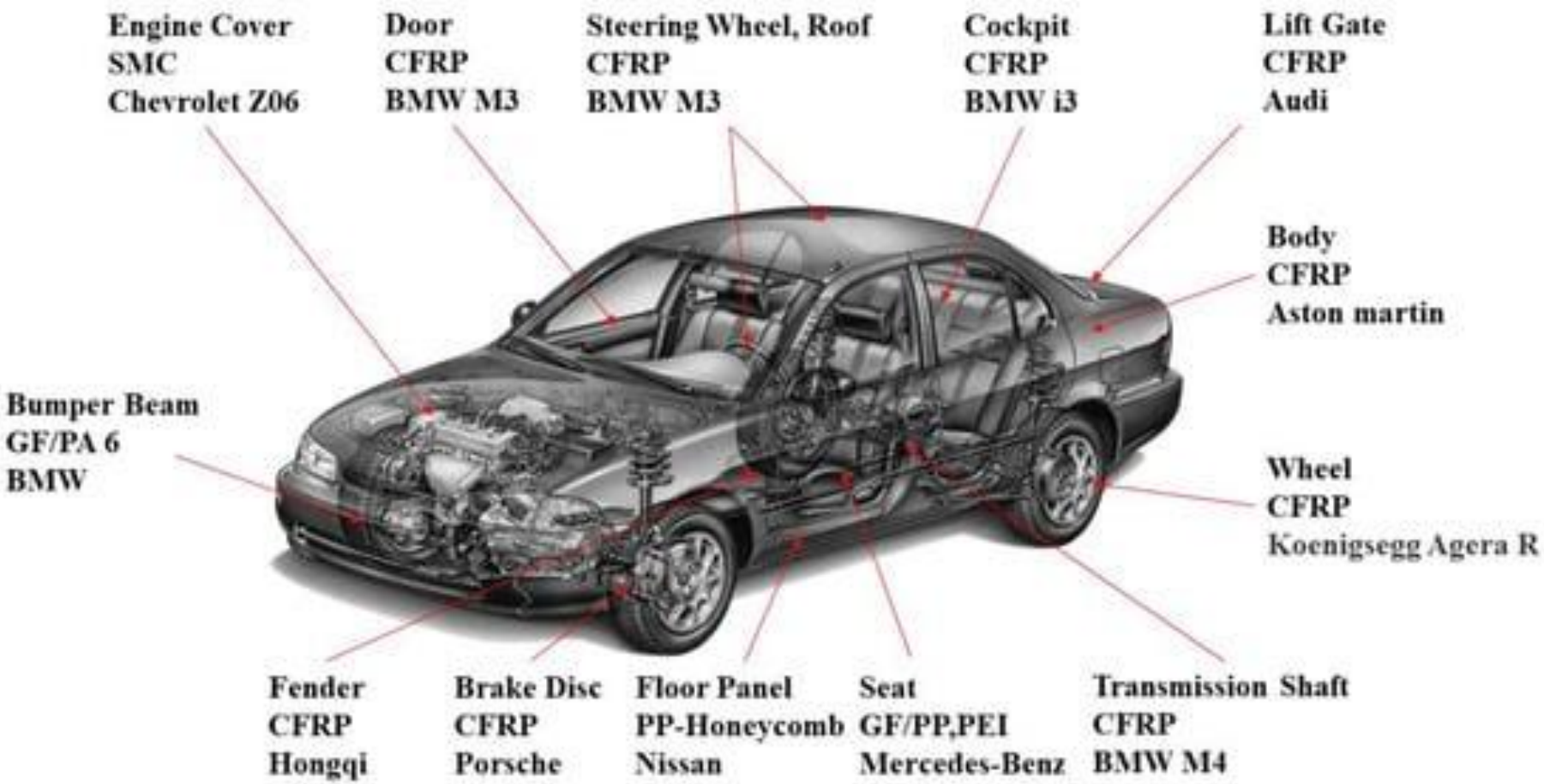
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Some more on composites in planes/cars, etc.



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Some more on composites in planes/cars, etc.



Du et al., Application of Lightweight Structure in Automobile Bumper Beam: A Review, Materials 2023, 16(3), 967; https://doi.org/10.3390/ma16030967